CAD MAINTAINED. CHANGES SHALL BE INCORPORATED BY THE CURRENT DESIGN ACTIVITY				
REVISION HISTORY – ALL SHEETS ARE THE SAME REVISION				
ZONE	REV	DESCRIPTION	DATE	APPROVED
	ATP1	PROTO BUILD	04–FEB–2025	

P-1

CERTIFICATION DRAWING
NO MODIFICATIONS PERMITTED WITHOUT APPROVAL OF THE APPROVAL ENGINEER AND REFERENCE TO THE CERTIFICATION AGENCIES.

REVISION STATUS

REV	RECORD OF CHANGE	DATE & NAME	REV	RECORD OF CHANGE	DATE & NAME
ATP1	PROTO BUILD	13APR2026 & Shyam Manikanta			

REISSUED

REVISION STATUS

REV	RECORD OF CHANGE	DATE & NAME	REV	RECORD OF CHANGE	DATE & NAME

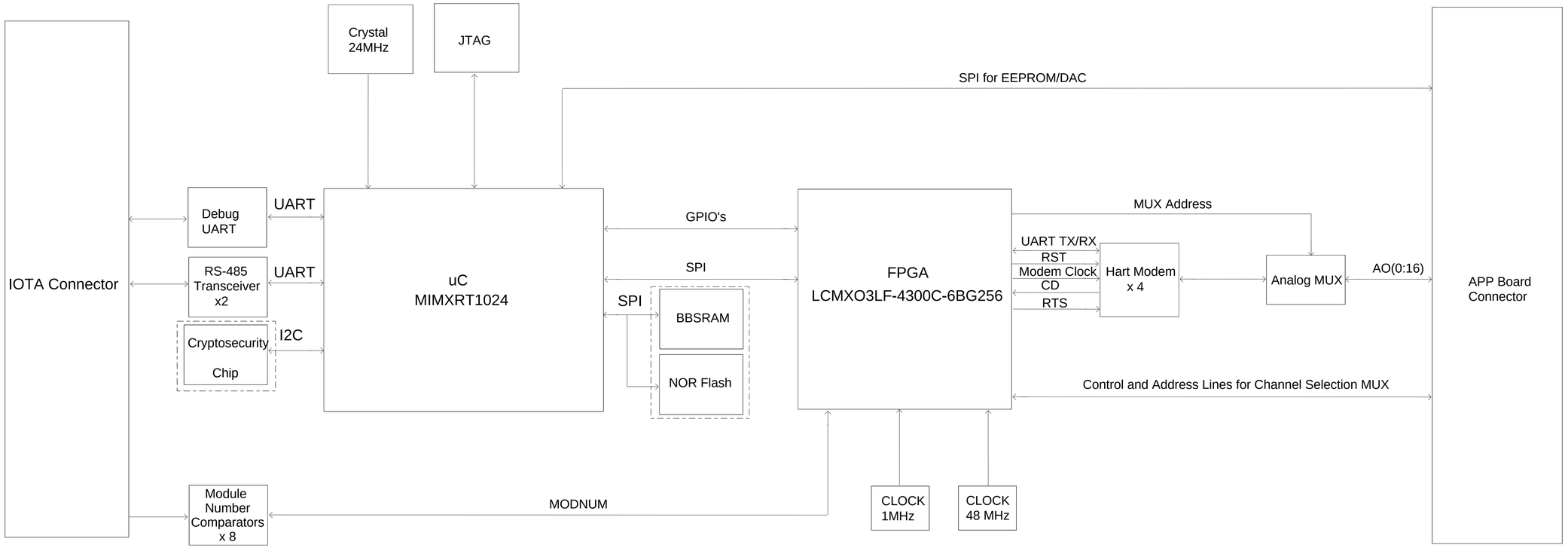
REISSUED

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xDxD VER:		HONEYWELL CONFIDENTIAL AND PROPRIETARY	HONEYWELL INTERNATIONAL INC. Honeywell
xDxD PATCH:			Honeywell Process Solutions
PWA #: 50195112	THIRD ANGLE PROJECTION	DRAWN BY Shyam Manikanta 24–OCT–2025	NAME AO IOM KERNEL BOARD
PWB #: 50195111		DESIGNER Shyam Manikanta 04–FEB–2025	
OS NUMBER: NA			SIZE D / DRAWING NO. 50195113 / REV ATP1
INITIAL APPLICATION			SCALE: 1:1 / WT: NA / SHEET 1 OF 22

P-2



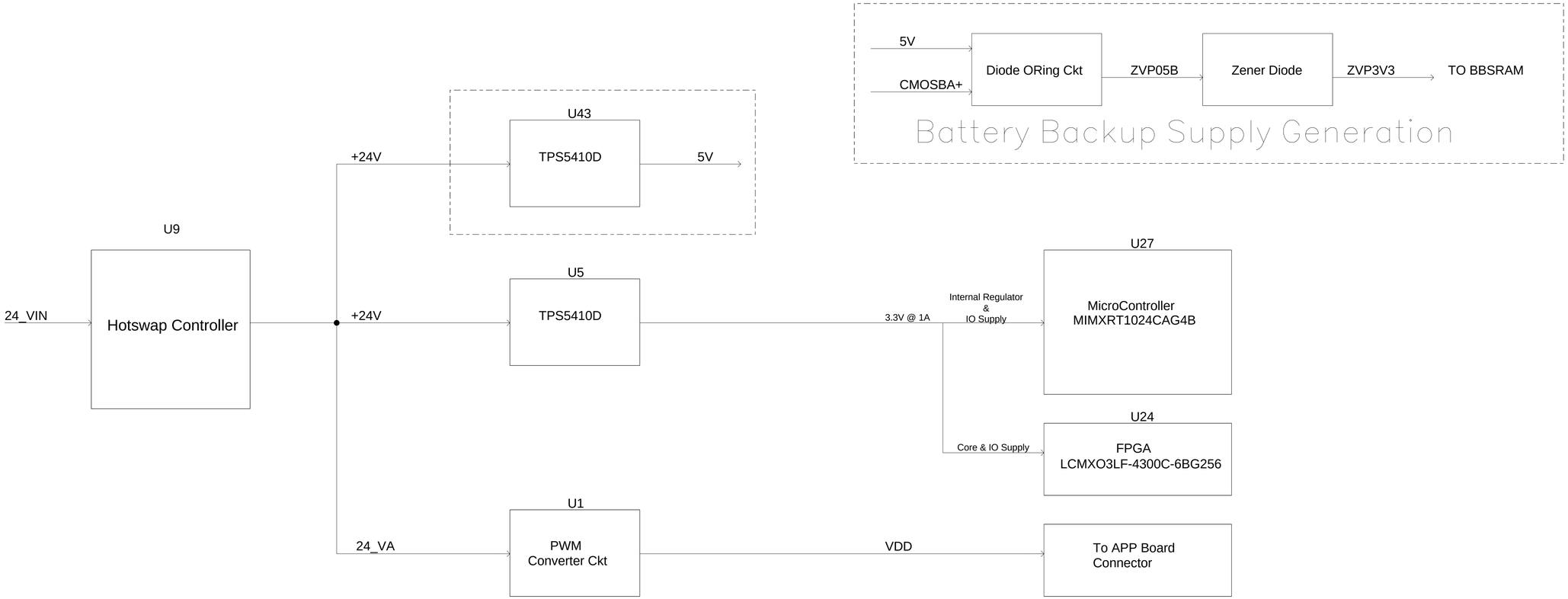
AO Board Logic Block

New Addition

DRAWN BY Shyam Manikanta	24-OCT-2025	TITLE	
DESIGNER Shyam Manikanta	24-OCT-2025	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions	
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D / DRAWING NO. 50195113 / REV XX	
		SCALE: 1:1	SHEET 2 OF 22

DxDESIGNER 10/04/2026:09:56

P-2



New Addition

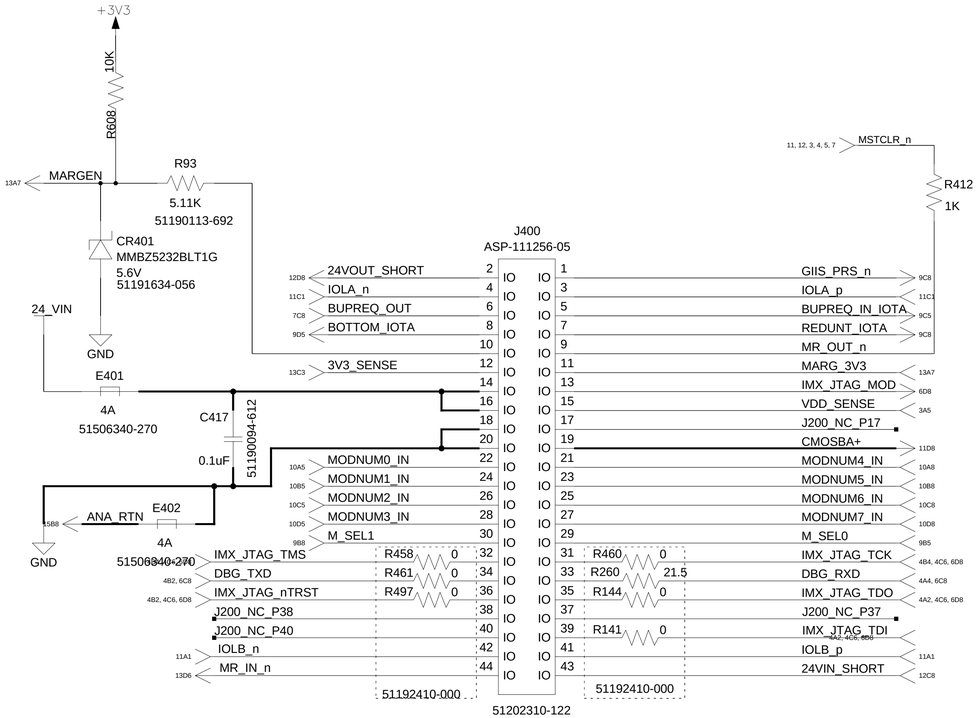
Only for PMIO

AO Board Power Generation Architecture

DRAWN BY Shyam Manikanta	24-OCT-2025	TITLE: AO Board Power Block	
DESIGNER Shyam Manikanta	24-OCT-2025	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions	
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D / DRAWING NO. 50195113 / REV ATP1	
		SCALE: 1:1	SHEET 2 OF 22

DxDESIGNER 07/04/2026:17:44

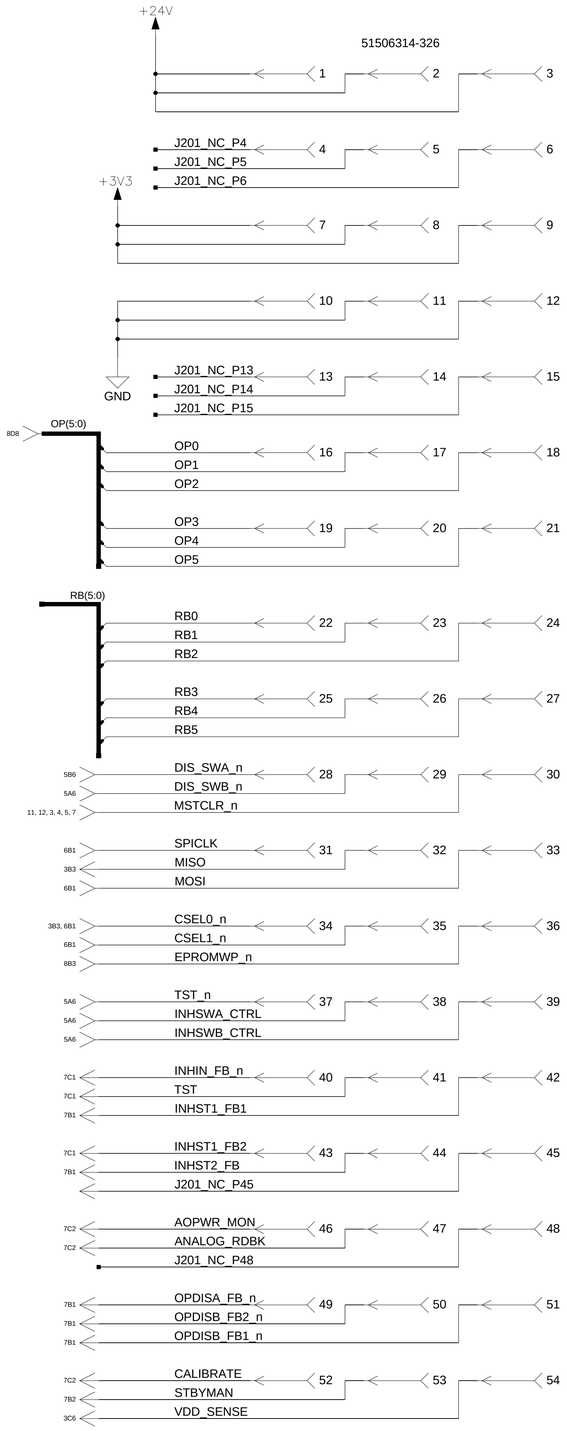
P-3

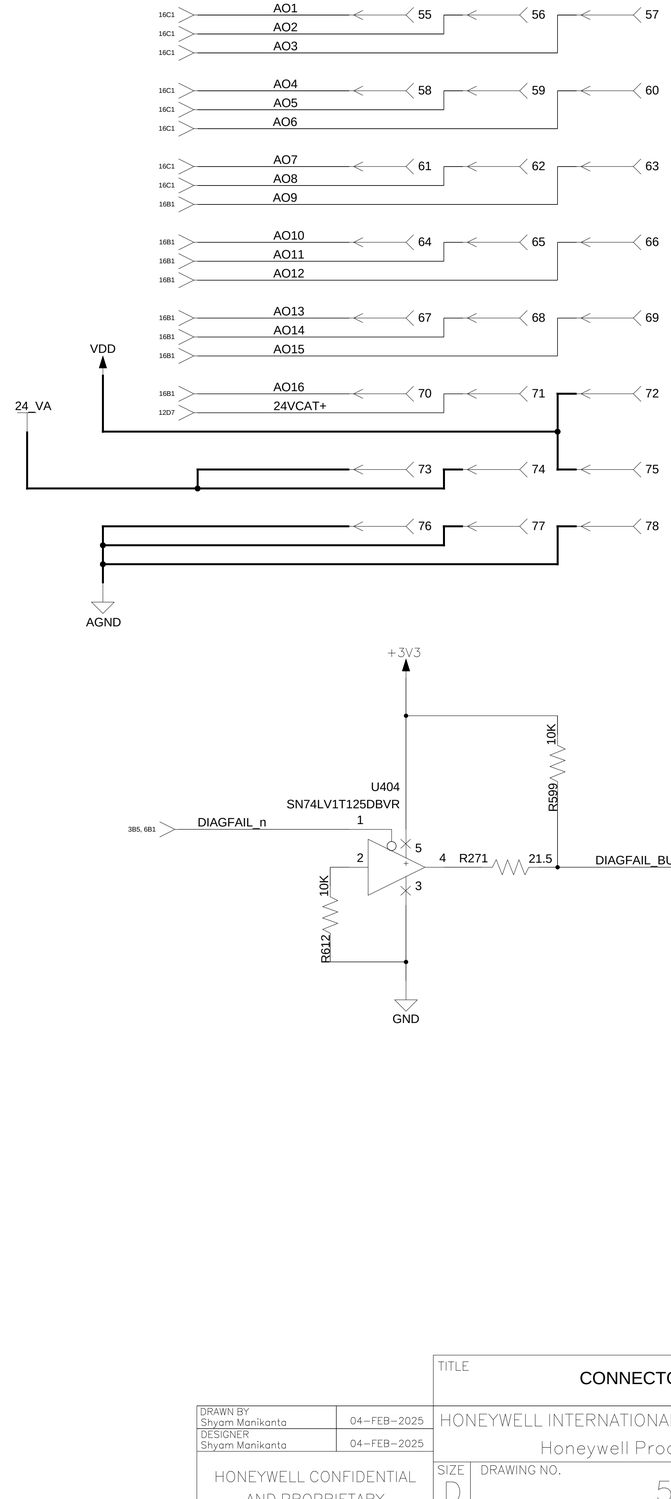


CONNECTOR JA Kernel to IOTA
Pins 2 and 43 are short pins on IOTA

Note: For PM AO:
Replace 51202310-122 with 51508516-344#MTLW-122-23-S-D-288

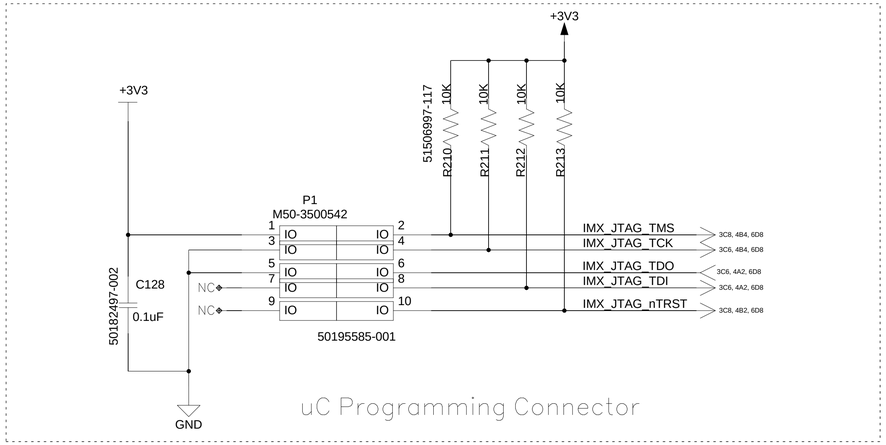
Note:
Removed Private path and added TDI & TDO Signals for uC JTAG

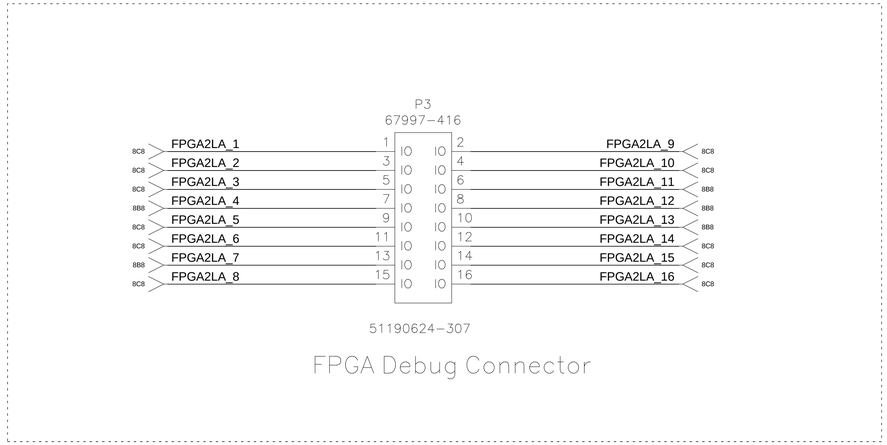


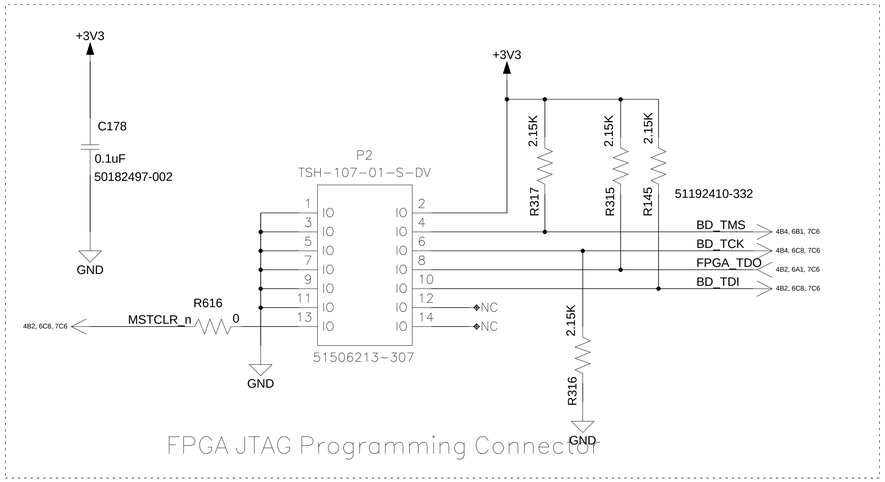


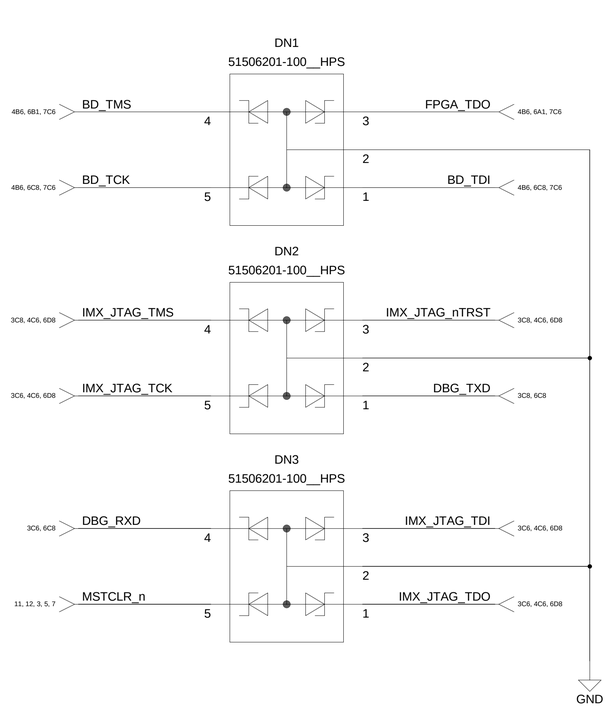
DRAWN BY Shyam Manikanta	04-FEB-2025	TITLE: CONNECTORS
DESIGNER Shyam Manikanta	04-FEB-2025	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D · DRAWING NO. 50195113 · REV ATP1
		SCALE: 1:1 · SHEET 3 OF 22

DxDESIGNER 16/04/2026:12:13





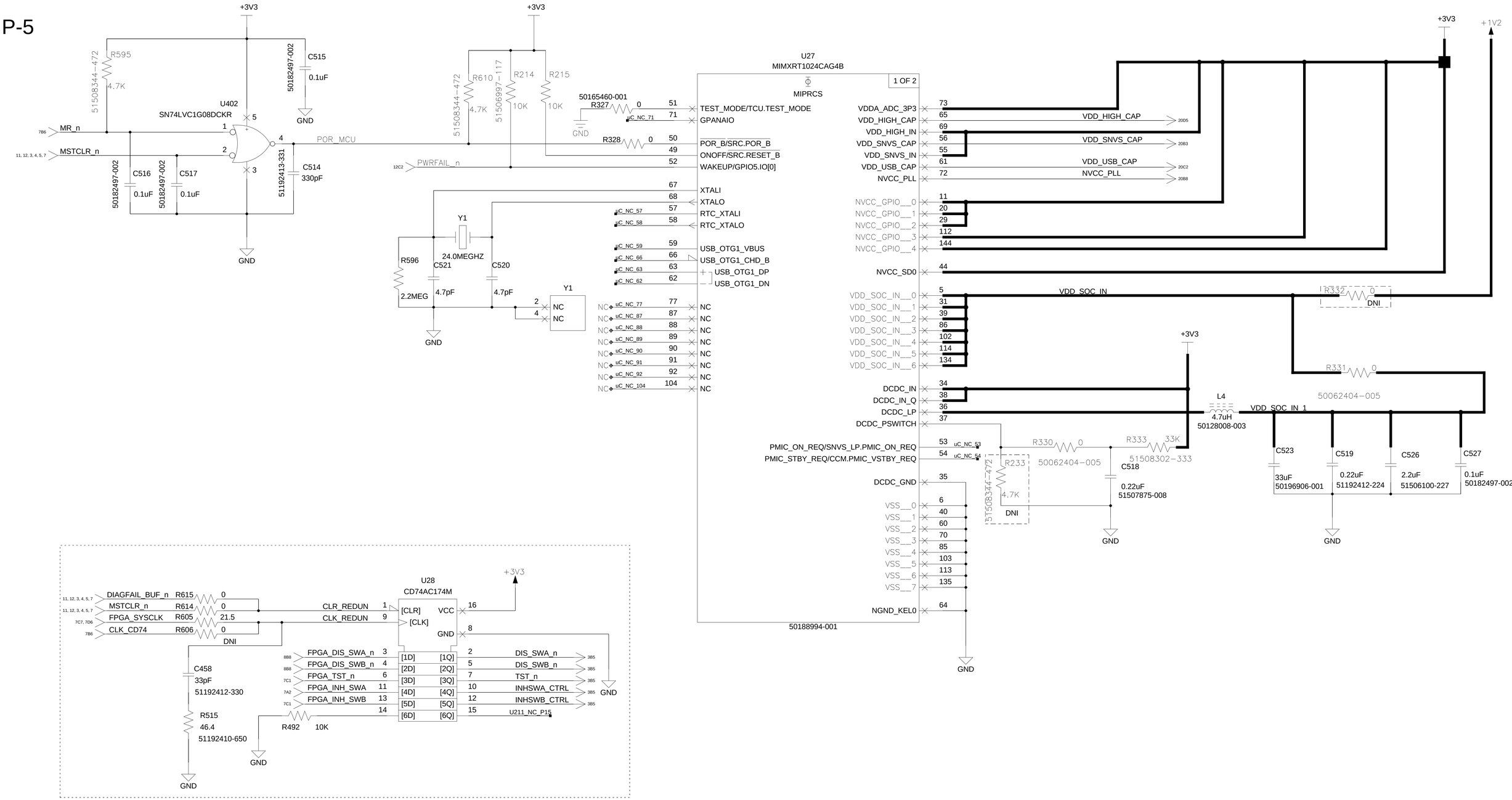




TITLE	uC,FPGA JTAG & FPGA Debug Connectors		
DRAWN BY Shyam Manikanta	04-FEB-2025	HONEYWELL INTERNATIONAL INC.	Honeywell
DESIGNER Shyam Manikanta	04-FEB-2025	Honeywell Process Solutions	
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D / DRAWING NO. 50195113 / REV ATP1	
		SCALE: 1:1	SHEET 4 OF 22

DxDESIGNER 13/04/2026:11:40

P-5



DRAWN BY Shyam Manikanta	04-FEB-2025	TITLE	uC Power
DESIGNER Shyam Manikanta	04-FEB-2025	HONEYWELL INTERNATIONAL INC.	Honeywell
HONEYWELL CONFIDENTIAL AND PROPRIETARY		Honeywell Process Solutions	
		SIZE D / DRAWING NO. 50195113 ATP1 / REV	
		SCALE: 1:1	SHEET 5 OF 22

DxDESIGNER 30/04/2026:16:39

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+3V3

51506997-117 R607 10K DNI
R185 10K
R184 10K
R183 10K
R182 10K
R181 10K
R177 10K
R176 10K
R149 10K
R148 10K
R147 10K
R146 10K

IMX_JTAG_TMS 111
IMX_JTAG_TCK 110
IMX_JTAG_MOD 109
R602 0 50165460-001
IMX_JTAG_TDI 108
IMX_JTAG_TDO 107
IMX_JTAG_nTRST 106
DBG_TXD R259 21.5 105
DBG_RXD 101
BD_TDI R258 21.5 100
PROGRAM_n R257 21.5 99
BD_TCK R234 21.5 98
WDTRFS_n R235 21.5 97
WDTODCLR_n R236 21.5 96
JABRES_n R237 21.5 95
IOL_TXD R262 21.5 94
IOL_RXD 93

TXREQ_n R272 21.5 84
RCVSELA_n R273 21.5 83
WDT_EN_n R274 21.5 82
BKP_CS1_n R256 21.5 81
REBOOT_n R275 21.5 80
SWBUREQ_n R276 21.5 79
BKP_SCLK R241 21.5 78
BKP_CS0_n R250 21.5 76
BKP_MOSI R244 21.5 75
BKP_MISO 74

R216 4.7K
GND

U27
MIMXRT1024CAG4B
2 OF 2
MIPRCS

GPIO_AD_B0_00/JTAG_MUX.TMS
GPIO_AD_B0_01/JTAG_MUX.TCK
GPIO_AD_B0_02/JTAG_MUX.MOD
GPIO_AD_B0_03/JTAG_MUX.TDI
GPIO_AD_B0_04/JTAG_MUX.TDO
GPIO_AD_B0_05/JTAG_MUX.TRSTB
GPIO_AD_B0_06/GPIO1.IO[6]
GPIO_AD_B0_07/GPIO1.IO[7]
GPIO_AD_B0_08/GPIO1.IO[8]
GPIO_AD_B0_09/GPIO1.IO[9]
GPIO_AD_B0_10/GPIO1.IO[10]
GPIO_AD_B0_11/GPIO1.IO[11]
GPIO_AD_B0_12/GPIO1.IO[12]
GPIO_AD_B0_13/GPIO1.IO[13]
GPIO_AD_B0_14/GPIO1.IO[14]
GPIO_AD_B0_15/GPIO1.IO[15]

GPIO_AD_B1_06/GPIO1.IO[22]
GPIO_AD_B1_07/GPIO1.IO[23]
GPIO_AD_B1_08/GPIO1.IO[24]
GPIO_AD_B1_09/GPIO1.IO[25]
GPIO_AD_B1_10/GPIO1.IO[26]
GPIO_AD_B1_11/GPIO1.IO[27]
GPIO_AD_B1_12/GPIO1.IO[28]
GPIO_AD_B1_13/GPIO1.IO[29]
GPIO_AD_B1_14/GPIO1.IO[30]
GPIO_AD_B1_15/GPIO1.IO[31]

50188994-001

51192410-330

GPIO_EMC_00/GPIO2.IO[0] 18 R280 21.5
GPIO_EMC_01/GPIO2.IO[1] 17 R281 21.5
GPIO_EMC_02/GPIO2.IO[2] 16 R282 21.5
GPIO_EMC_03/GPIO2.IO[3] 15 R283 21.5
GPIO_EMC_04/GPIO2.IO[4] 14 R284 21.5
GPIO_EMC_05/GPIO2.IO[5] 13 R285 21.5
GPIO_EMC_06/GPIO2.IO[6] 12 R286 21.5
GPIO_EMC_07/GPIO2.IO[7] 10 R287 21.5
GPIO_EMC_08/GPIO2.IO[8] 9 R586 21.5
GPIO_EMC_09/GPIO2.IO[9] 8 R587 21.5
GPIO_EMC_10/GPIO2.IO[10] 7 R588 21.5
GPIO_EMC_11/GPIO2.IO[11] 4 R589 21.5
GPIO_EMC_12/GPIO2.IO[12] 3 R590 21.5
GPIO_EMC_13/GPIO2.IO[13] 2 R591 21.5
GPIO_EMC_14/GPIO2.IO[14] 1 R592 21.5
GPIO_EMC_15/GPIO2.IO[15] 143 R593 21.5
GPIO_EMC_16/SRC_BOOT_MODE0 142
GPIO_EMC_17/SRC_BOOT_MODE1 141
GPIO_EMC_18/GPIO2.IO[18] 140
GPIO_EMC_19/GPIO2.IO[19] 139
GPIO_EMC_20/GPIO2.IO[20] 138
GPIO_EMC_21/GPIO2.IO[21] 137
GPIO_EMC_22/GPIO2.IO[22] 136
GPIO_EMC_23/GPIO2.IO[23] 133
GPIO_EMC_24/GPIO2.IO[24] 132
GPIO_EMC_25/GPIO2.IO[25] 131
GPIO_EMC_26/GPIO2.IO[26] 130
GPIO_EMC_27/GPIO2.IO[27] 129
GPIO_EMC_28/GPIO2.IO[28] 128 R291 21.5
GPIO_EMC_29/GPIO2.IO[29] 127 R288 21.5
GPIO_EMC_30/GPIO2.IO[30] 126 R289 21.5
GPIO_EMC_31/GPIO2.IO[31] 125 R290 21.5
GPIO_EMC_32/GPIO3.IO[0] 124 R269 21.5
GPIO_EMC_33/GPIO3.IO[1] 123 R267 21.5
GPIO_EMC_34/GPIO3.IO[2] 122 R270 21.5
GPIO_EMC_35/GPIO3.IO[3] 121
GPIO_EMC_36/GPIO3.IO[4] 120 R268 21.5
GPIO_EMC_37/GPIO3.IO[5] 119
GPIO_EMC_38/GPIO3.IO[6] 118
GPIO_EMC_39/GPIO3.IO[7] 117
GPIO_EMC_40/GPIO3.IO[8] 116
GPIO_EMC_41/GPIO3.IO[9] 115

GPIO_SD_B0_00/GPIO3.IO[13] 48 R584 21.5
GPIO_SD_B0_01/GPIO3.IO[14] 47 R585 21.5
GPIO_SD_B0_02/GPIO3.IO[15] 46 R240 21.5
GPIO_SD_B0_03/GPIO3.IO[16] 45 R238 21.5
GPIO_SD_B0_04/GPIO3.IO[17] 43
GPIO_SD_B0_05/GPIO3.IO[18] 42
GPIO_SD_B0_06/GPIO3.IO[19] 41 R277 21.5

GPIO_SD_B1_00/GPIO3.IO[20] 33 R581 21.5
GPIO_SD_B1_01/GPIO3.IO[21] 32 R580 21.5
GPIO_SD_B1_02/GPIO3.IO[22] 30
GPIO_SD_B1_03/GPIO3.IO[23] 28
GPIO_SD_B1_04/GPIO3.IO[24] 27 R278 21.5
GPIO_SD_B1_05/GPIO3.IO[25] 26 R279 21.5
GPIO_SD_B1_06/GPIO3.IO[26] 25 R263 21.5
GPIO_SD_B1_07/GPIO3.IO[27] 24 R264 21.5
GPIO_SD_B1_08/GPIO3.IO[28] 23 R265 21.5
GPIO_SD_B1_09/GPIO3.IO[29] 22
GPIO_SD_B1_10/GPIO3.IO[30] 21
GPIO_SD_B1_11/GPIO3.IO[31] 19

51508344-472 R217 4.7K
R218 4.7K
R221 4.7K
R222 4.7K
R223 4.7K
R224 4.7K
R225 4.7K
R226 4.7K

R170 10K
R171 10K
R172 10K
R173 10K
R174 10K
51506997-117

GND

TP1
GND

+3V3

51506997-117 R201 10K
R200 10K
R199 10K
R198 10K
R197 10K
R196 10K
R195 10K
R186 10K
R178 10K
R179 10K
R180 10K
R175 10K
R150 10K
R313 1K
R314 1K
R168 10K
R169 10K
51508344-472 R227 4.7K
R228 4.7K
R229 4.7K
R230 4.7K
51506997-117 R202 10K
R203 10K
R204 10K
R205 10K
R206 10K
R207 10K
R208 10K
R209 10K

ADDR(0) 7B2
ADDR(1) 7B2
ADDR(2) 7B2
ADDR(3) 7B2
ADDR(4) 7B2
ADDR(5) 7B2
ADDR(6) 7B2
ADDR(7) 7B2
DATA[0] 7B2
DATA[1] 7B2
DATA[2] 7B2
DATA[3] 7B2
DATA[4] 7B2
DATA[5] 7B2
DATA[6] 7B2
DATA[7] 7B2
MD0 7A2
MD1 6A7, 9B2
BT_CFG[0] 6A7, 9B6
BT_CFG[1] 6B7
BT_CFG[2] 6B7
BT_CFG[3] 6B7
BT_CFG[4] 6A7
BT_CFG[5]
BT_CFG[6]
BT_CFG[7]
BT_CFG[8] 6A7
BT_CFG[9] 6A7
CS_n
AS_n 7C6
RD_n 7C6
WR_n 7C6
SPICLK 7C6
CSEL0_n 3B5
MOSI 3B3, 3B5
MISO 3B5
CSEL1_n 3B2
BUPREQ_FB_FPGA 3B5
BUPREQ_IN_FPGA 8B8
BOTTOM_FPGA 8B8
REDUNT_FPGA 8B8
OVERVT_n_FPGA 8B8

TPM_SCL 8B8
TPM_SDA 22C7
DIAGFAIL_n 22C7
CLRGHL_n 7C6
GHOIRQ_n 7D6
FPGA_IRQ 7C6
BD_TMS 7D6
4B4, 4B6, 7C6

INIT_n 7A5
FPGA_DONE 7A6
FPGA_TDO 4B2, 4B6, 7C6
ANALOG_RDBK_FPGA 7C2
WAIT_n 7B6
IMX_SPARE_1 7C2
IMX_FPGA_CS0_n 7B6
IMX_FPGA_SCLK 8B8
IMX_FPGA_MOSI 7B6
IMX_FPGA_MISO 7B6
JABLOCK 8B8
GHOXXX 8B8

+3V3

R308 1K
R309 1K
R295 1K
R296 1K
R297 1K
R298 1K
R299 1K
R294 1K DNI
51190131-012

BT_CFG[0] 6C1
BT_CFG[1] 6C1
BT_CFG[2] 6C1
BT_CFG[3] 6C1
BT_CFG[8] 6C1
BT_CFG[9] 6C1
MD0 6C1, 9B2
MD1 6C1, 9B6

R306 1K DNI
R307 1K
R300 1K
R301 1K
R302 1K
R303 1K
R304 1K
R305 1K
51190131-012

GND

MD1	MD0	BOOT TARGET
0	0	BOOT FROM FUSES
0	1	SERIAL DOWNLOAD
1	0	INTERNAL BOOT

BOOT MODE & BOOT CONFIG SEL

TITLE: iMXRT1024 Microcontroller

DRAWN BY Shyam Manikanta	04-FEB-2025
DESIGNER Shyam Manikanta	04-FEB-2025

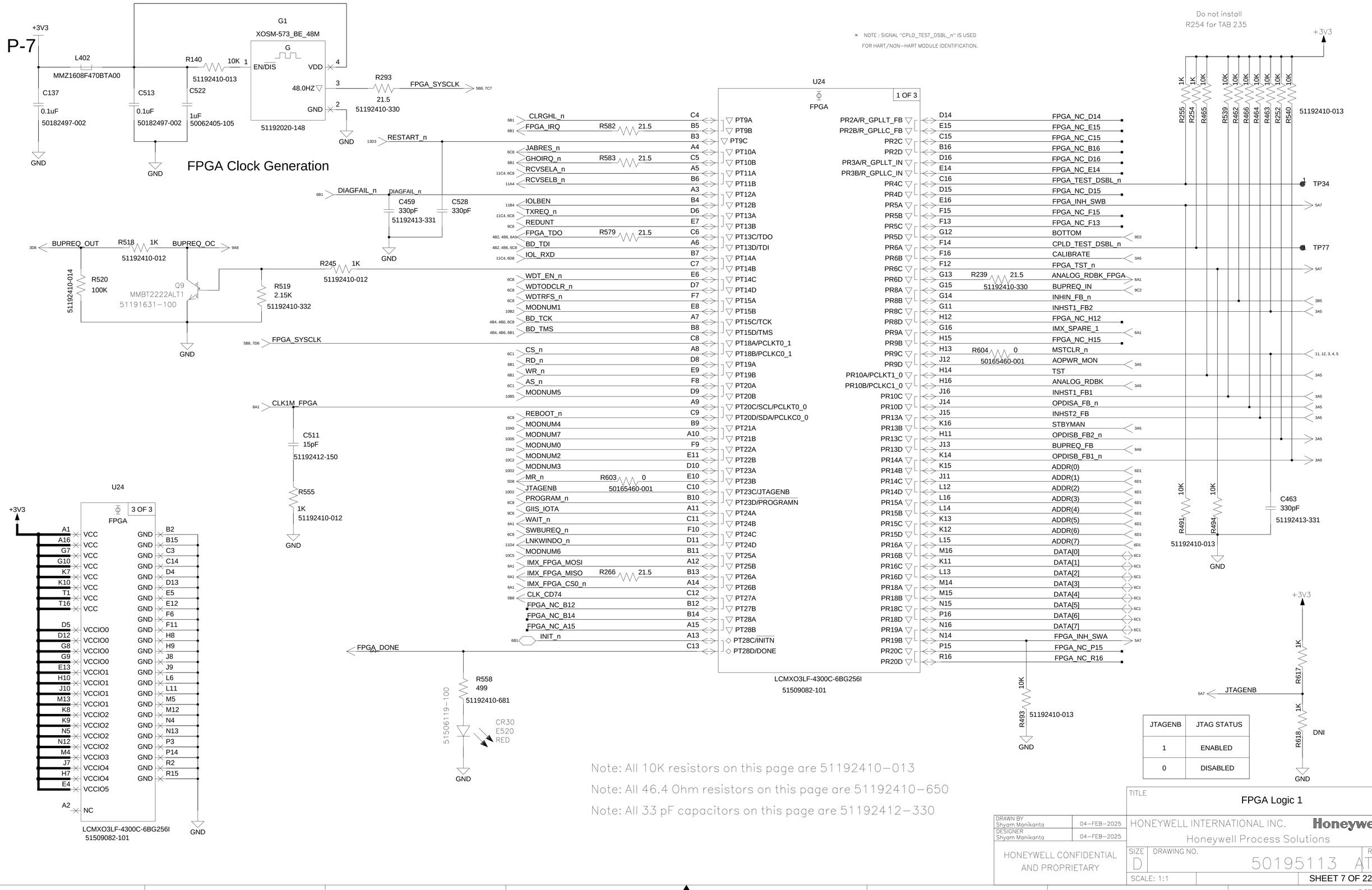
HONEYWELL CONFIDENTIAL AND PROPRIETARY

HONEYWELL INTERNATIONAL INC. Honeywell
Honeywell Process Solutions

SIZE D | DRAWING NO. 50195113 | REV ATP1

SCALE: 1:1 | SHEET 6 OF 22

DxDESIGNER 16/04/2026:12:13


P-7
FPGA Clock Generation
+3V3
L402
MMZ1608F470BTA00
C137
0.1uF
50182497-002
C513
0.1uF
50182497-002
R140
10K
51192410-013
C522
1uF
50062405-105
G1
XOSM-573_BE_48M
EN/DIS
VDD
48.0HZ
GND
51192020-148
R293
21.5
51192410-330
FPGA_SYSCLK
RESTART_n
DIAGFAIL_n
C459
330pF
51192413-331
C528
330pF
BUPREQ_OUT
R518
1K
51192410-012
BUPREQ_OC
R520
100K
51192410-014
Q9
MMBT2222ALT1
51191631-100
R519
2.15K
51192410-332
R245
1K
51192410-012
CLK1M_FPGA
C511
15pF
51192412-150
R555
1K
51192410-012
U24
FPGA
3 OF 3
LCMXO3LF-4300C-6BG256I
51509082-101
CLRGHL_n
FPGA_IRQ
R582
21.5
JABRES_n
GHOIRQ_n
R583
21.5
RCVSELA_n
RCVSELB_n
IOLBEN
TXREQ_n
REDUNT
FPGA_TDO
R579
21.5
BD_TDI
IOL_RXD
WDT_EN_n
WDTODCLR_n
WDTRFS_n
MODNUM1
BD_TCK
BD_TMS
CS_n
RD_n
WR_n
AS_n
MODNUM5
REBOOT_n
MODNUM4
MODNUM7
MODNUM0
MODNUM2
MODNUM3
MR_n
R603
0
50165460-001
JTAGENB
PROGRAM_n
GIIS_IOTA
WAIT_n
SWBUREQ_n
LNKWINDO_n
MODNUM6
IMX_FPGA_MOSI
IMX_FPGA_MISO
R266
21.5
IMX_FPGA_CS0_n
CLK_CD74
FPGA_NC_B12
FPGA_NC_B14
FPGA_NC_A15
INIT_n
FPGA_DONE
R558
499
51192410-681
CR30
E520
RED
51506119-100
U24
FPGA
1 OF 3
LCMXO3LF-4300C-6BG256I
51509082-101
* NOTE : SIGNAL "CPLD_TEST_DSBL_n" IS USED FOR HART/NON-HART MODULE IDENTIFICATION.
FPGA_NC_D14
FPGA_NC_E15
FPGA_NC_C15
FPGA_NC_B16
FPGA_NC_D16
FPGA_NC_E14
FPGA_TEST_DSBL_n
FPGA_NC_D15
FPGA_INH_SWB
FPGA_NC_F15
FPGA_NC_F13
BOTTOM
CPLD_TEST_DSBL_n
CALIBRATE
FPGA_TST_n
R239
21.5
51192410-330
ANALOG_RDBK_FPGA
BUPREQ_IN
INHIN_FB_n
INHST1_FB2
FPGA_NC_H12
IMX_SPARE_1
FPGA_NC_H15
R604
0
50165460-001
MSTCLR_n
AOPWR_MON
TST
ANALOG_RDBK
INHST1_FB1
OPDISA_FB_n
INHST2_FB
STBYMAN
OPDISB_FB2_n
BUPREQ_FB
OPDISB_FB1_n
ADDR(0)
ADDR(1)
ADDR(2)
ADDR(3)
ADDR(4)
ADDR(5)
ADDR(6)
ADDR(7)
DATA[0]
DATA[1]
DATA[2]
DATA[3]
DATA[4]
DATA[5]
DATA[6]
DATA[7]
FPGA_INH_SWA
FPGA_NC_P15
FPGA_NC_R16
R493
10K
51192410-013
Do not install
R254 for TAB 235
R255
1K
R254
1K
R465
10K
R539
10K
R462
10K
R466
10K
R464
10K
R463
10K
R252
10K
R540
10K
51192410-013
TP34
TP77
R491
10K
R494
10K
51192410-013
C463
330pF
51192413-331
JTAGENB
R617
1K
R618
1K
DNI
JTAGENB
JTAG STATUS
1
ENABLED
0
DISABLED
Note: All 10K resistors on this page are 51192410-013
Note: All 46.4 Ohm resistors on this page are 51192410-650
Note: All 33 pF capacitors on this page are 51192412-330
TITLE
FPGA Logic 1
DRAWN BY Shyam Manikanta 04-FEB-2025
DESIGNER Shyam Manikanta 04-FEB-2025
HONEYWELL CONFIDENTIAL AND PROPRIETARY
HONEYWELL INTERNATIONAL INC.
Honeywell Process Solutions
SIZE D
DRAWING NO. 50195113
REV ATP1
SCALE: 1:1
SHEET 7 OF 22
DxDESIGNER 30/04/2026:16:36

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U24
LCMXO3LF-4300C-6BG256I
2 OF 3
FPGA

OP(5:0) ← 3C5
RB(5:0)

OP0	P4	PB3A	
OP1	T4	PB3B	
OP2	T2	PB3C	
RB(0)	R3	PB3D	
OP3	R5	PB4A/CSSPIN	
FPGA_NC_R5	P5	PB4B	
OP4	T3	PB6A	
OP5	R4	PB6B	
RB(2)	T5	PB7A	
4C2 FPGA2LA_14	R6	PB7B	
RB(1)	P6	PB9A/MCLK/CCLK	
RB(3)	T6	PB9B/SO/SPISO	
4C2 FPGA2LA_15	N6	PB9C	
RB(4)	L7	PB9D	
RB(5)	R7	PB10A	
4C4 FPGA2LA_8	P7	PB10B	
4C2 FPGA2LA_16	M7	PB10C	
FPGA_NC_N7	N7	PB10D	
4D2 FPGA2LA_9	M6	PB12A	
4C4 FPGA2LA_5	L8	PB12B	
4D4 FPGA2LA_1	T7	PB13A/PCLKT2_0	
FPGA_NC_R8	R8	PB13B/PCLKC2_0	
FPGA_NC_P8	P8	PB15A	
4D4 FPGA2LA_2	T8	PB15B	
4D2 FPGA2LA_10	N8	PB15C	
4D4 FPGA2LA_3	L9	PB15D	
4C4 FPGA2LA_6	M8	PB18A	
4D2 FPGA2LA_11	N9	PB18B	
6A1 IMX_FPGA_SCLK	T9	PB20A/PCLKT2_1	
5B7 FPGA_DIS_SWA_n	P9	PB20B/PCLKC2_1	
4D2 FPGA2LA_12	R9	PB21A	
4D4 FPGA2LA_4	T10	PB21B	
4C4 FPGA2LA_7	M9	PB21C	
5A7 FPGA_DIS_SWB_n	L10	PB21D	
4C2 FPGA2LA_13	P10	PB23A	
FPGA_NC_R10	R10	PB23B	
6B1 BUPREQ_FB_FPGA	N10	PB23C	
6B1 BUPREQ_IN_FPGA	M11	PB23D	
6B1 BOTTOM_FPGA	T11	PB24A	
6B1 REDUNT_FPGA	P11	PB24B	
6B1 OVERVT_n_FPGA	M10	PB24C	
6A1 JABLOCK	N11	PB24D	
6A1 GHOXXX	R13	PB26A	
FPGA_NC_T14	T14	PB26B	
FPGA_NC_R11	R11	PB27A	
FPGA_NC_T12	T12	PB27B	
FPGA_NC_P12	P12	PB29A	
FPGA_NC_T13	T13	PB29B	
FPGA_NC_R12	R12	PB30A/SN	
FPGA_NC_P13	P13	PB30B/SI/SISPI	
FPGA_NC_T15	T15	PB30C	
FPGA_NC_R14	R14	PB30D	

R577 21.5
R578 21.5

51192410-013
10K R496
10K R495
GND

51509082-101

+3V3
10K R514
10K R251
10K R536
10K R242
10K R537
10K R538
10K R534
10K R551
51192410-013

PL16A	L1	MER(1)	
PL16B	L3	MER(0)	
PL16C	K4	HARTRST0_n	17C7
PL16D	L5	MER(3)	
PL17A/PCLKT3_0	L2	MER(2)	
PL17B/PCLKC3_0	M1	MAR(0)	
PL17C	K5	HART0TXD	17B4
PL17D	L4	MAR(1)	
PL18C	N2	HART0RTS_n	17D5
PL18D	P1	MAR(6)	
PL19A	M3	MAR(2)	
PL19B	N1	MAR(3)	
PL20A	M2	MAR(5)	
PL20B	N3	HART1TXD	17A3
PL20C	R1	HART1RTS_n	17B5
PL20D	P2	HARTRST1_n	17A6
PL9A	G1	MAR(7)	
PL9B	H2	MAR(4)	
PL9C	H4	MAR(10)	
PL9D	J6	MAR(9)	
PL10A	H3	MAR(8)	
PL10B	H1	MAR(12)	
PL10C/PCLKT4_0	J1	MAR(11)	
PL10D/PCLKC4_0	J3	MAR(13)	
PL13A	J2	MAR(14)	
PL13B	K1	HART2TXD	18C4
PL13C	H5	HARTRST2_n	18C7
PL13D	J4	HART2RTS_n	18D5
PL14A	K3	MAR(15)	
PL14B	K2	HART3TXD	18A4
PL14C	J5	HARTRST3_n	18A6 → 3B5
PL14D	K6	EPROMWP_n	TP70
PL2C	B1	FPGA_NC_B2	
PL2D	C2	LEDGRN_n	8D3
PL3A/L_GPLLT_FB	D3	LEDRED_n	8D1
PL3B/L_GPLLC_FB	D1	FPGA_NC_D1	
PL4A/L_GPLLT_IN	E2	FPGA_NC_E2	
PL4B/L_GPLLC_IN	E3	HART3RTS_n	18B5
PL4C	C1	HART0CD	17C4
PL4D	D2	HART0RXD	17C4
PL6A/PCLKT5_0	E1	HART1CD	17B4
PL6B/PCLKC5_0	F2	MODEMCLK	17, 18
PL6C	F4	HART1RXD	17A4
PL6D	G6	HART3RXD	18A4
PL7A	F3	HART2RXD	18C4
PL7B	F1	HART2CD	18C4
PL7C	G5	HART3CD	18B4
PL7D	G4	FPGA_NC_G4	
PL8A	G2	FPGA_NC_G2	
PL8B	G3	FPGA_NC_G3	
PL8C	F5	FPGA_NC_F5	
PL8D	H6	FPGA_NC_H6	

R535 21.5
51192410-330

10K R243
10K R249
10K R531
10K R248
GND

MER(3:0) ← 16
MAR(15:0) ← 16

+3V3
R21 4.64K 51192410-652
R23 4.64K 51192410-652
LEDGRN_n ← 8B4
R20 1K 51192410-012
Q3 MMBT2907A 51191617-101
Q4 MMBT2907A 51191617-101
R22 1K 51192410-012 → LEDRED_n 8B4
R4 100
R5 100
D400
RED
GREEN
XLUGR29M 5V
GND
+3V3
R2 100 51192410-011
XLUGR29M
GREEN
5V
RED
D1
TP21
GND

Note: FOR PM AO
MAKE R2 &D1 AS DNP

+3V3
51192410-441 R561 280
R560 280 51192410-441
G2
51506287-121__HPS
G
1.0MEGHZ
V+
GND
51190094-612
C94 0.1uF
R601 21.5 51192410-330 → CLK1M_FPGA 7B7
GND

Note: For PM AO:
Replace 51506323-100 with 51506313-104#TW-03-09-S-S-180-100

TITLE	FPGA LOGIC 2	
DRAWN BY Shyam Manikanta	04-FEB-2025	HONEYWELL INTERNATIONAL INC. Honeywell
DESIGNER Shyam Manikanta	04-FEB-2025	Honeywell Process Solutions
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D · DRAWING NO. 50195113 · REV ATP1
		SCALE: 1:1 · SHEET 8 OF 22

DxDESIGNER 10/04/2026:11:31

P-9

+24V
R117 0
51192410-000

R415
0

+3V3
51192410-013
R125 10K

+24V
51192410-183
R84 15K 63mW

U17
LM2901D

3D6 GIIS_PRS_n
R85 1K
51192410-012
CR18
51191634-133
51192413-331
C409 330pF
GND

4 3
5 2
12
51190379-102

GIIS_IOTA 7B6

+3V3
51192410-013
R88 10K

+24V
R119 15K

3D6 REDUNT_IOTA
R120 1K
51192410-012
CR17
51191634-133
51192413-331
C408 330pF
GND

U17
LM2901D
6 3
7 1
12

REDUNT 7C6

51192410-000
R570 0
DNI

+3V3
51192410-013
R73 10K DNI

+24V
R33 15K

U7
LM2901D

3C8 M_SEL1
R35 1K
51192410-012
CR6
51191634-133
51192413-331
C22 330pF
GND

6 3
7 1
12

51192410-000
R571 0
DNI

MD1 6A7, 6C1

+3V3
51192410-013
R72 10K

+24V
*

7C7 BUPREQ_OC
R53 10K
51192410-013
CR7
51191634-133
51192413-331
C34 330pF
GND

U7
LM2901D
4 3
5 2
12

BUPREQ_FB 7B2

+24V
51192410-692
R31 5.11K

REF_17V9 10D7, 13C4

51190094-612
C20 0.1uF

R30 15K
51192410-183
GND

+24V
51192410-183
R118 15K

3D8 BOTTOM_IOTA
R121 1K
51192410-012
CR24
51191634-133
51192413-331
C414 330pF
GND

U17
LM2901D
8 3
9 14
12

+3V3
51192410-013
R122 10K

BOTTOM 7C2

+24V
R151 15K
51192410-183

+24V
*

+3V3
51192410-013
R123 10K

3D6 BUPREQ_IN_IOTA
CR29
51191634-133
R152 1K
51192410-012
CR28
51191634-133
51192413-331
C421 330pF
GND

U17
LM2901D
10 3
11 13
12

BUPREQ_IN 7C2

51192410-000
R572 0

+24V
R66 15K
51192410-183

+24V
+3V3

3C6 M_SEL0
R65 1K
51192410-012
CR5
51191634-133
51192413-331
C31 330pF
GND

U7
LM2901D
10 3
11 13
12

51192410-000
R573 0
DNI

MD0 6A7, 6C1

+3V3
51192410-013
R71 10K

+3V3
3V3by2

+24V
+3V3

51192410-013
R50 10K

+24V
51192410-013
R70 10K

3D8 MARGEN
R29 1K
51192410-012
MARGIN_R

U7
LM2901D
8 3
9 14
12

51192410-000
R600 0

MARGEN_IN 12A8

51190379-102

51192410-013
R28 10K

51190094-612
C32 0.1uF

51192410-013
R51 10K

C33 0.1uF 10% 50V

+3V3
U405
5
2 4
3
74AHC1G04GW,125
DNI
GND

U405
74AHC1G04GW,125
U405_NC_1 1
NC

NOTE: Gates marked with * must be in different packages to isolate single points of failure

NOTE: For Series C: Place R600, DNI U405
For PMIO: DNI R600, Place U405

DRAWN BY Shyam Manikanta	04-FEB-2025
DESIGNER Shyam Manikanta	04-FEB-2025

HONEYWELL CONFIDENTIAL AND PROPRIETARY

TITLE: IOTA Config Comparators

HONEYWELL INTERNATIONAL INC.
Honeywell Process Solutions

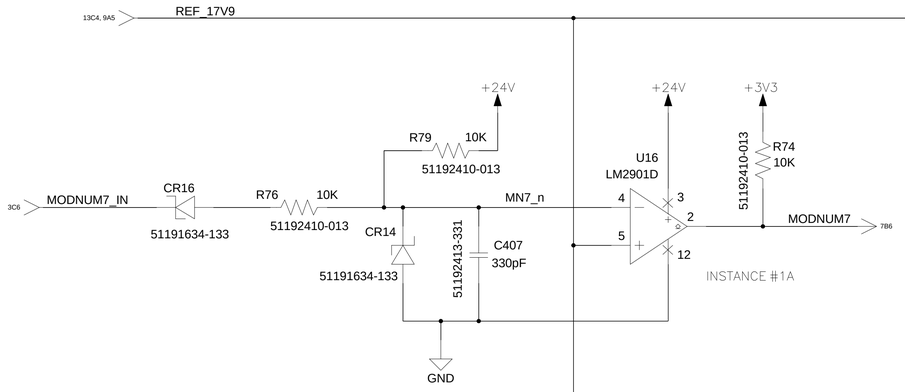
SIZE D | DRAWING NO. 50195113 | REV ATP1

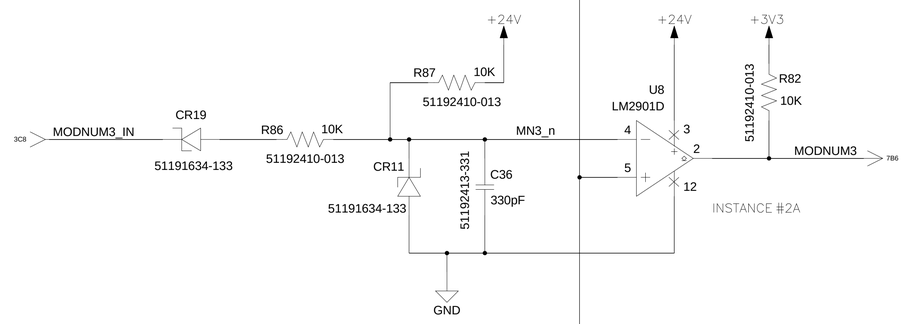
SCALE: 1:1

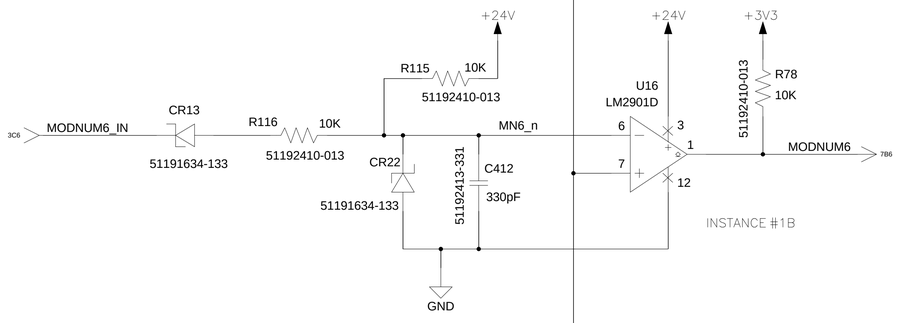
SHEET 9 OF 22

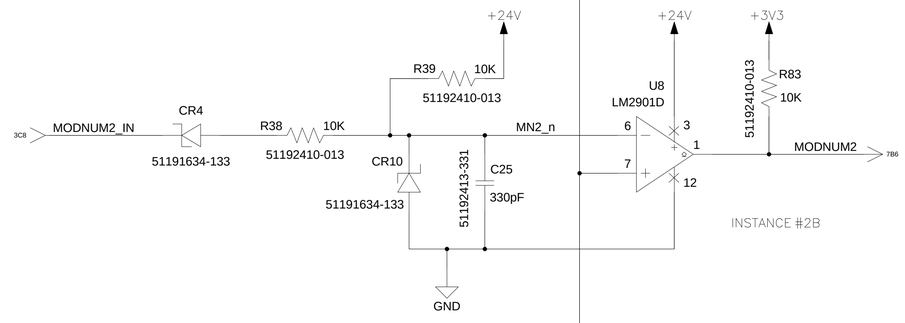
DxDESIGNER 10/04/2026:11:36

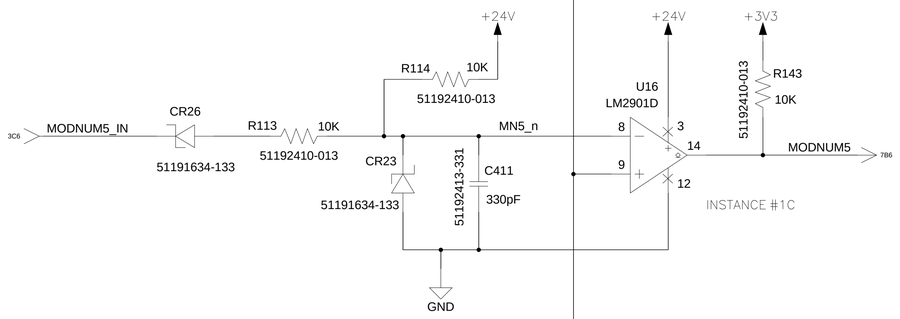
P-10

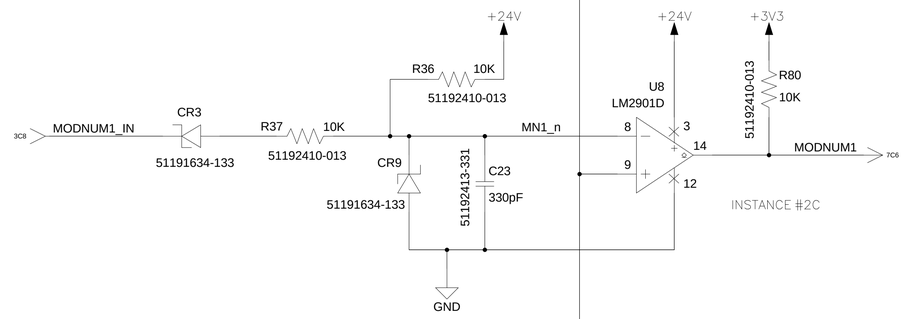


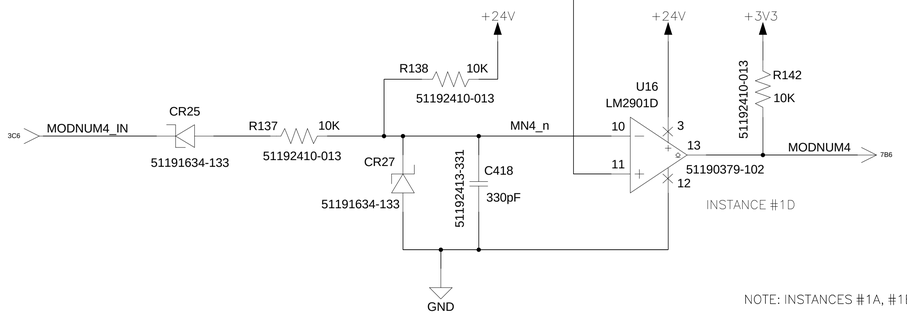


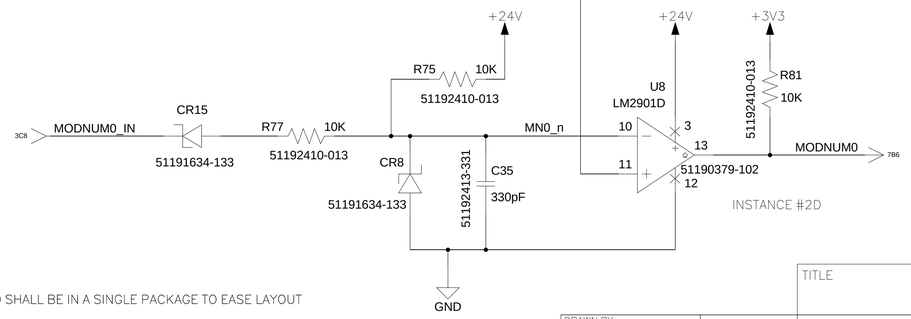












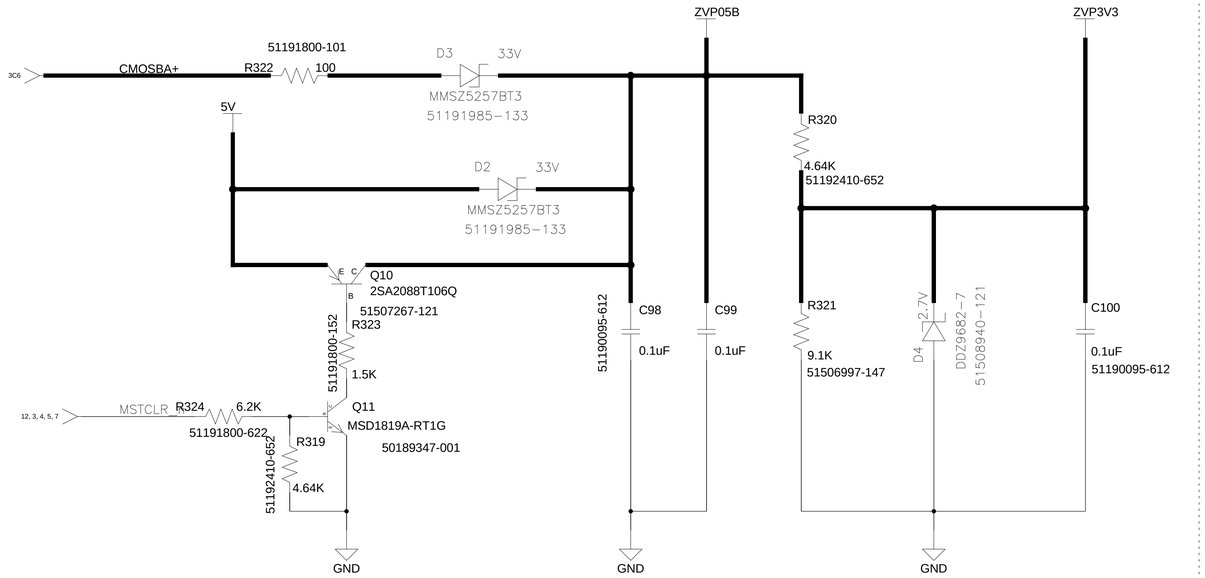
NOTE: INSTANCES #1A, #1B, #1C, #1D SHALL BE IN A SINGLE PACKAGE TO EASE LAYOUT

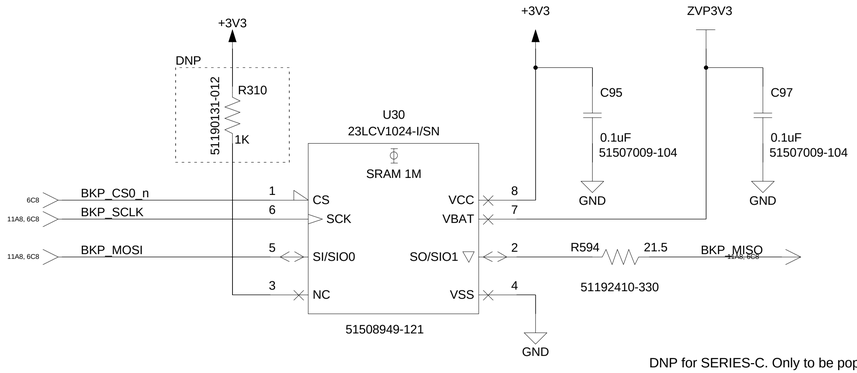
NOTE: INSTANCES #2A, #2B, #2C, #2D SHALL BE IN A SINGLE PACKAGE TO EASE LAYOUT

TITLE	Module Number Comparators
DRAWN BY Shyam Manikanta	04-FEB-2025
DESIGNER Shyam Manikanta	04-FEB-2025
HONEYWELL CONFIDENTIAL AND PROPRIETARY	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions
SIZE D	DRAWING NO. 50195113 ATP1
SCALE: 1:1	SHEET 10 OF 22

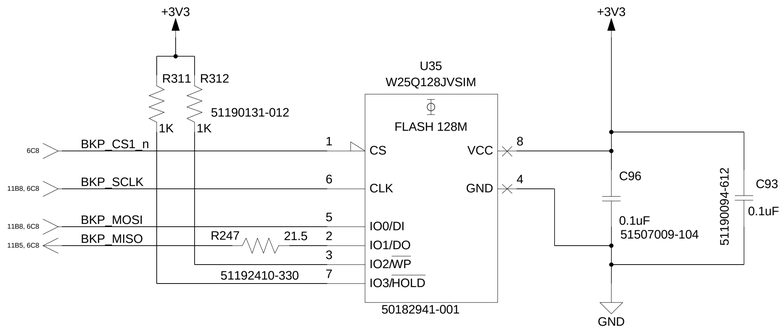
DxDESIGNER 31/03/2026:16:45

P-11

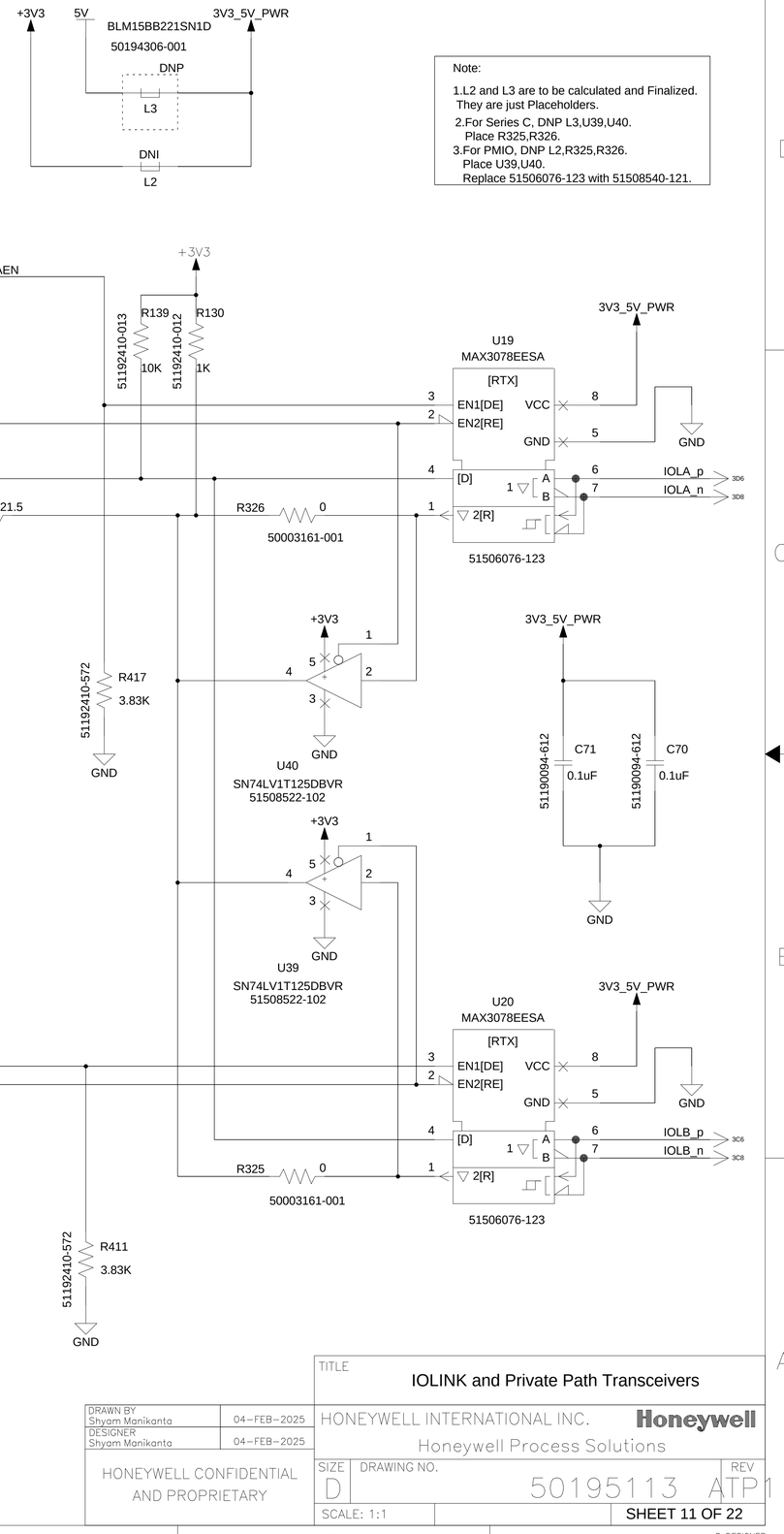




DNP for SERIES-C. Only to be populated for PMIO



NOR Flash for Failure Diagnostics



TITLE	IOLINK and Private Path Transceivers	
DRAWN BY Shyam Manikanta	04-FEB-2025	HONEYWELL INTERNATIONAL INC. Honeywell
DESIGNER Shyam Manikanta	04-FEB-2025	Honeywell Process Solutions
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D · DRAWING NO. 50195113 · REV ATP1
		SCALE: 1:1 · SHEET 11 OF 22

D:\DESIGNER 01/04/2026:18:25

24_VIN

CR21
MRA4005T1
600V

24VCAT+ 3C3

3D8 24VOUT_SHORT R92 5.11K
51190113-692

Connected by IOTA

3C6 24VIN_SHORT

R42
3.16K
51190113-492

R40
1K
51190131-012

C27
0.01uF
51190094-496

GND

Hot Swap Turn–On Threshold
Low to High – 11.91V min, 13.36V max
High to Low – 11.67V min, 12.94V max

Refer to LTC1641 Datasheet for Layout Instructions
Trip Current 1.06A min, 1.31A typ, 1.57A max

U10
SI4850EY

R91 36m
51506134-036

SW_GATE

51506192–121

R41
10
51192410-010

U9
LT1641–1CS8
HOT SWAP CONTR
ON
FB
SENSE
VCC
TIMER
PWRGD
GATE
GND

51506099–125

HS_TIMER
HS_GATE

C50
0.1uF
51190094-612

C48
0.68uF
51506100-068

C49 0.01uF HSN10
51190094-496

R90
1K
51190131-012

CR12
18V
51191634-118

GND

+24V
24_VA
W1
NAMCHG
x_no_pn_namchg_smt_020

+3V3

R56
10K
51192410-013

R569 10K
51192410-013

PWRFAIL_n 5C6

PWRFAIL Threshold (High to Low) = 15.97V to 17.40V
(Low to High) = 16.4V to 18.2V

C529
330pF
51192413-331

GND

R55
60.4K
51190131-763

R54
5.11K
51190113-692

GND

+3V3

R562
10K
51192410-013

D402
70V
51506258-100

R598
3.16K
51192410-492

DNI

R564
10K
51192410-013

Q12
MMBT2907A
51191617-101

R563
10K
51192410-013

3V3_OV

U403
VOLT MON
VCC
MR
SET
GND
OV
UV

MAX6759UTTD3PLUST
51506281-121

9A2 MARGEN_IN

3V3_UVR576 0 MSTCLR_n
51192410-000

C512
0.1uF
51507009-104

GND

RESET THRESHOLD (Normal)

OverVoltage Threshold:
3.465V min, 3.548V typ, 3.630V max

UnderVoltage Threshold:
2.97V min, 3.053V typ, 3.135V max

RESET THRESHOLD (Margining)

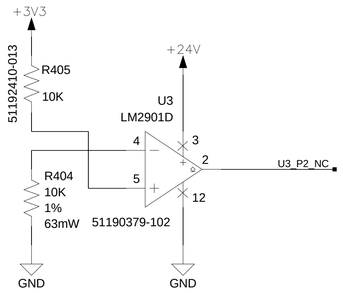
OverVoltage Threshold:
3.630V min, 3.713V typ, 3.795V max

UnderVoltage Threshold:
2.805V min, 2.888V typ, 2.970V max

DRAWN BY Shyam Manikanta	04–FEB–2025	TITLE: Hot Swap Controller, Voltage Supervisor
DESIGNER Shyam Manikanta	04–FEB–2025	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D · DRAWING NO. 50195113 · REV ATP1
		SCALE: 1:1 · SHEET 12 OF 22

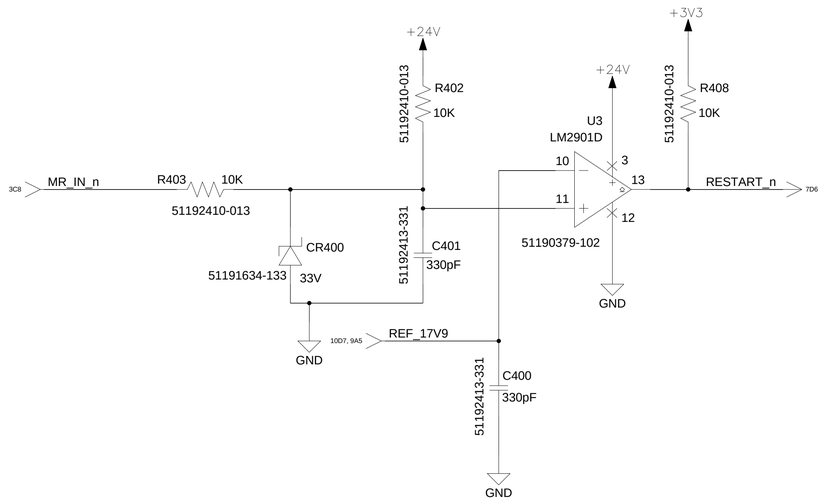
DxDESIGNER 16/04/2026:11:59

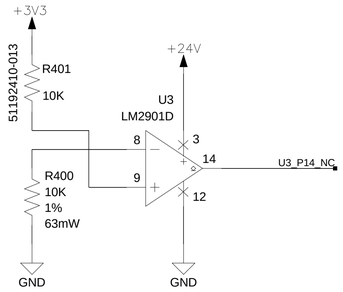
P-13

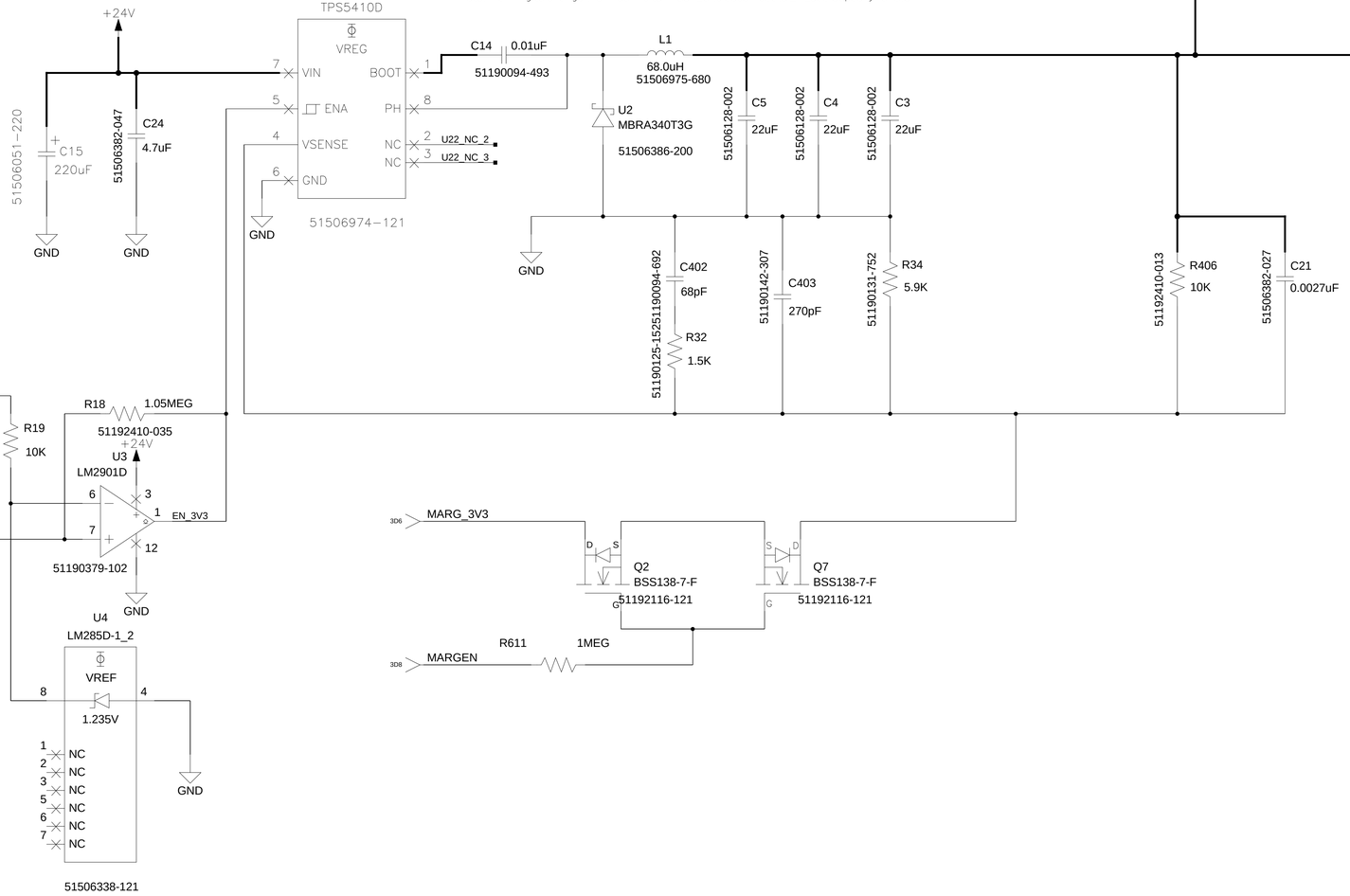


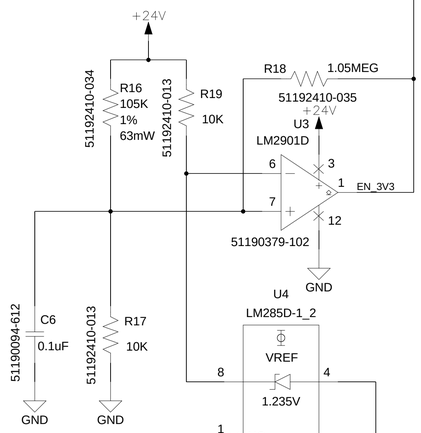
3.3V @ 1A max
3.21V min to 3.44V max
Load 300 mA max

Refer to TPS5410 datasheet for layout guidelines





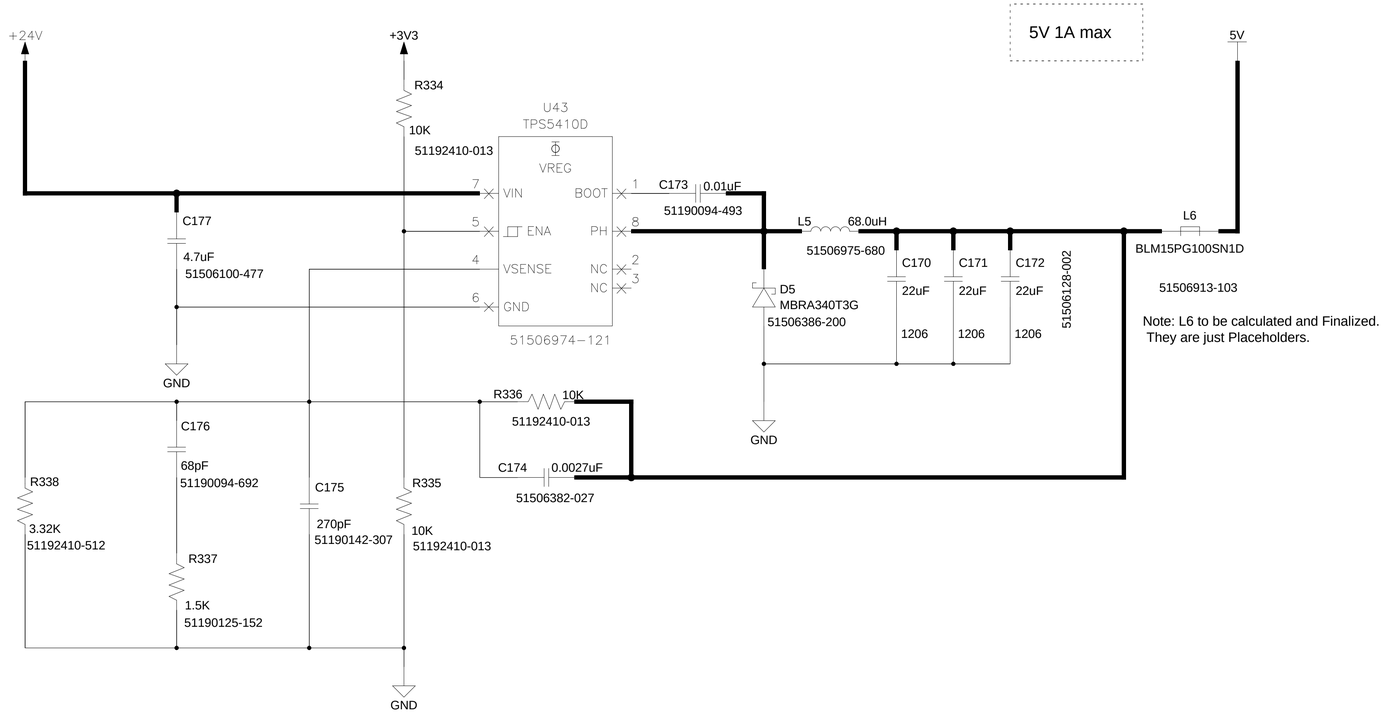




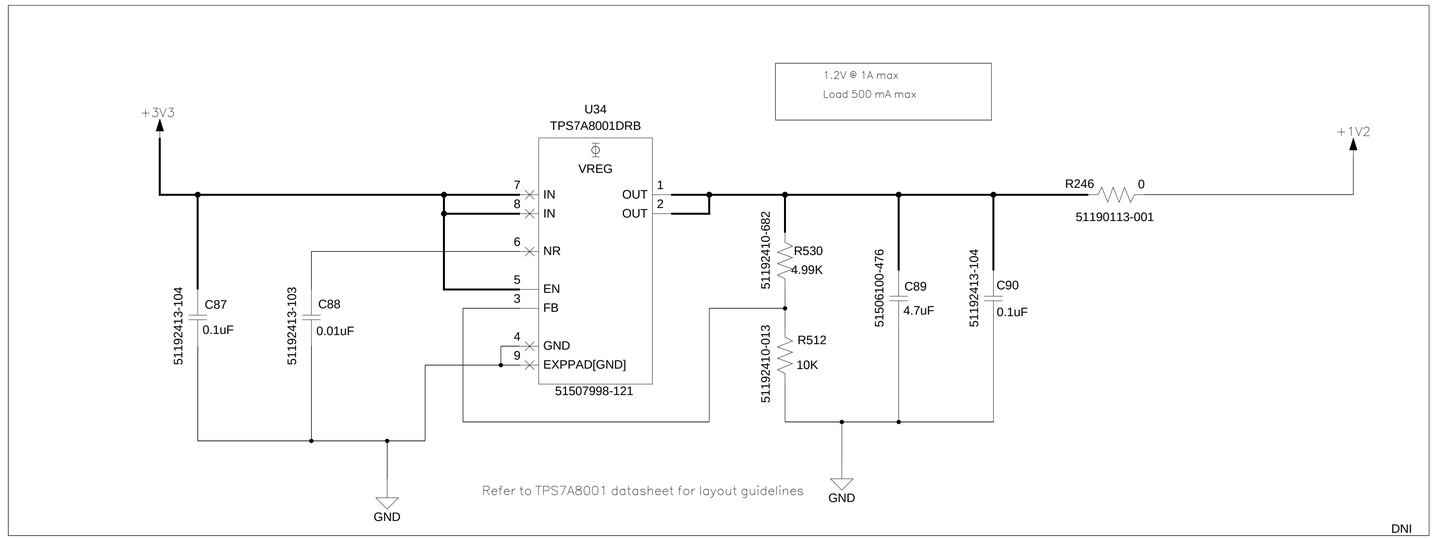
DRAWN BY Shyam Manikanta	04-FEB-2025	TITLE: 3V3 Supply Generation	
DESIGNER XXXXXXXX	04-FEB-2025	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions	
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D	DRAWING NO. 50195113 ATP1 REV
		SCALE: 1:1	SHEET 13 OF 22

DxDESIGNER 16/04/2026:12:02

P-14



This Regulator and the Passives are only to be populated for PM IOP



DRAWN BY Shyam Manikanta	04-FEB-2025
DESIGNER Shyam Manikanta	04-FEB-2025

HONEYWELL CONFIDENTIAL AND PROPRIETARY

TITLE: Linear Regulators for 5V Supply

HONEYWELL INTERNATIONAL INC. Honeywell

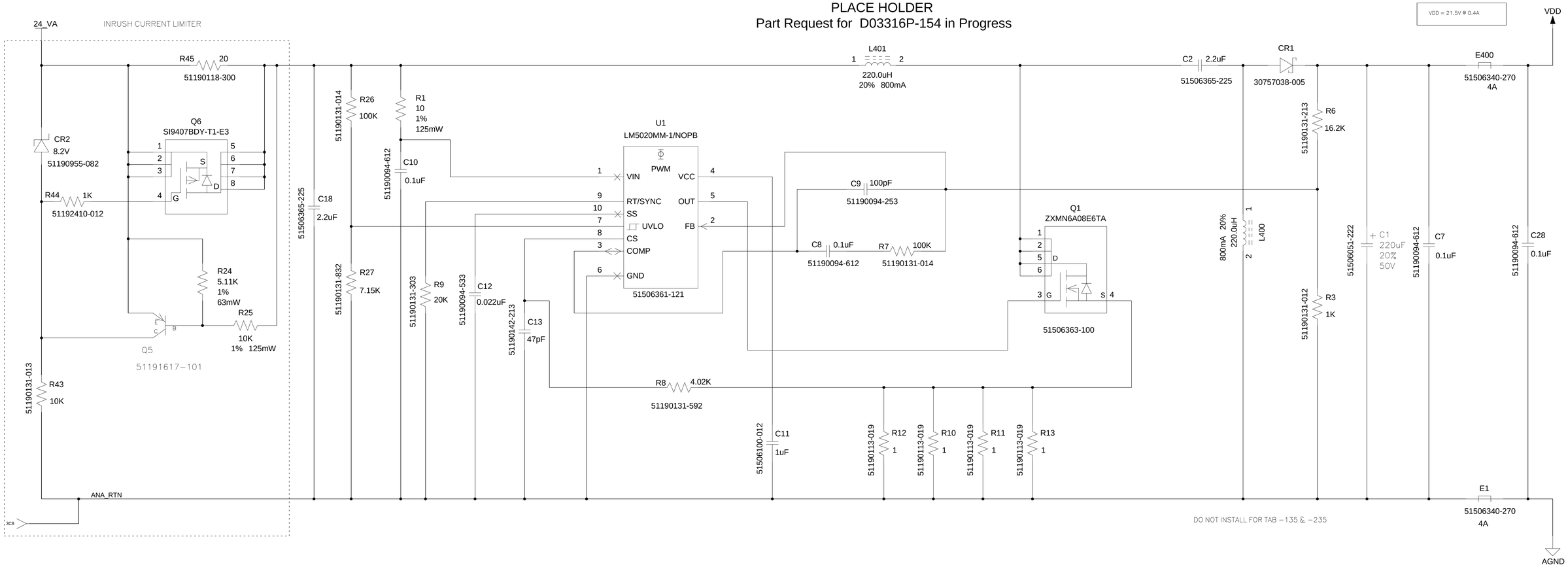
Honeywell Process Solutions

SIZE D | DRAWING NO. 50195113 | REV ATP1

SCALE: 1:1 | SHEET 14 OF 22

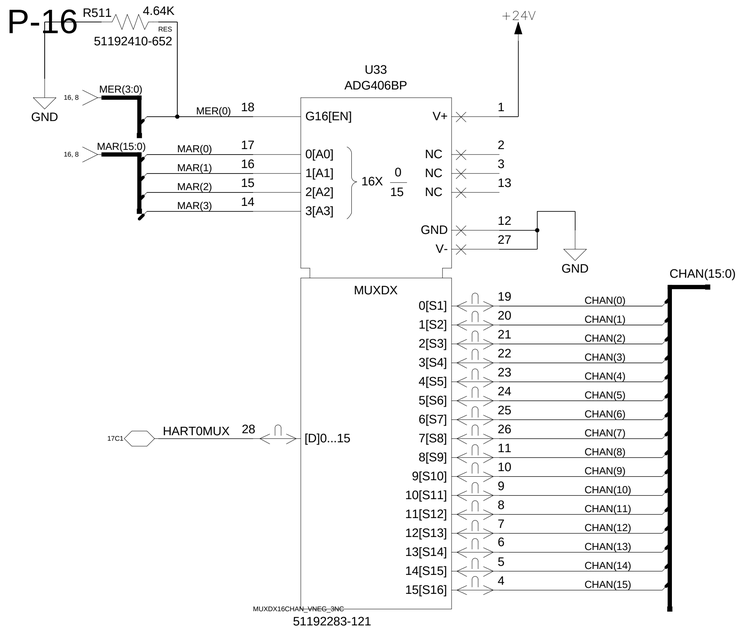
DxDESIGNER 10/04/2026:11:31

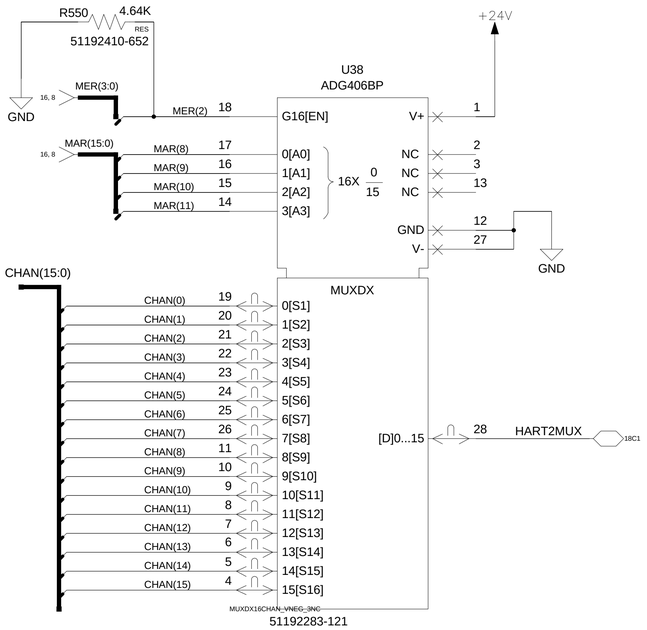
PLACE HOLDER
Part Request for D03316P-154 in Progress

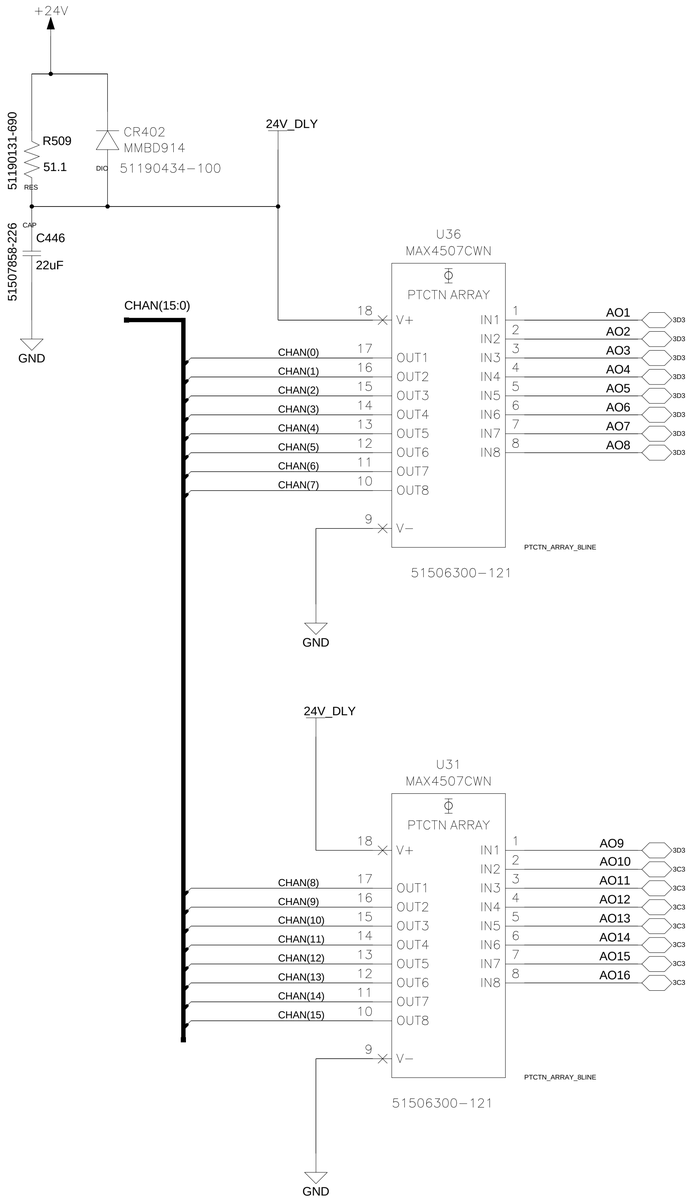


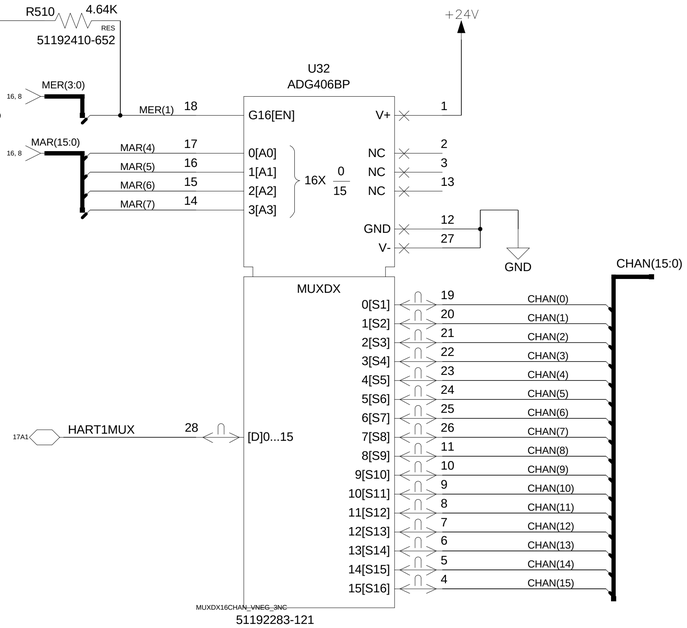
DRAWN BY Shyam Manikanta	04–FEB–2025	TITLE: Analog Supply Generation
DESIGNER Shyam Manikanta	04–FEB–2025	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D · DRAWING NO. 50195113 · REV ATP1
		SCALE: 1:1 · SHEET 15 OF 22

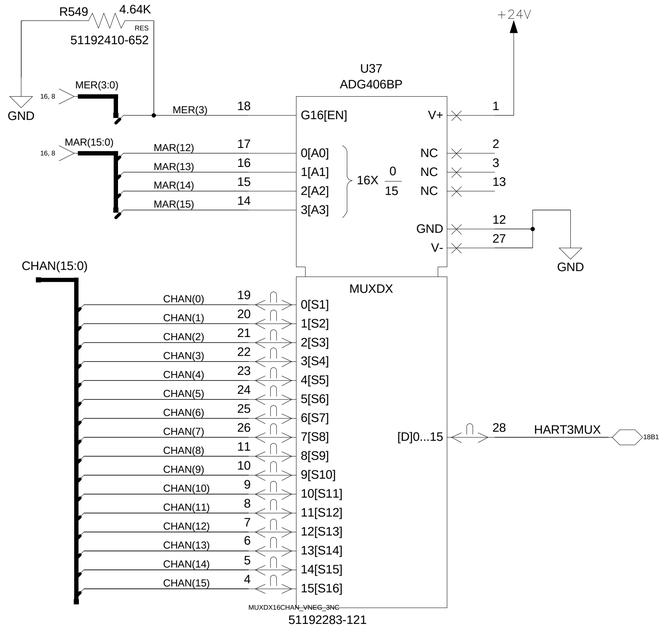
DxDESIGNER 07/05/2026:12:00











ALL COMPONENTS ON THIS SHEET ARE NOT INSTALLED FOR TAB –235

DRAWN BY Shyam Manikanta	04–FEB–2025	TITLE: Analog Multiplexers
DESIGNER Shyam Manikanta	04–FEB–2025	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D · DRAWING NO. 50195113 · REV ATP1
		SCALE: 1:1 · SHEET 16 OF 22

DxDESIGNER 10/04/2026:11:31

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MODEM 0

U14
A5191HRTNG-XTP
MODEM
51507504-121

U12
LM285D-1_2
VREF
1.235V
51506338-121

U6
OP2177AR
OPAMP
51506189-121

U11
MAX312LESE
SW_ANLG_SPST_NOCONN
51506339-121

ALL COMPONENTS ON THIS SHEET ARE NOT INSTALLED FOR TAB –235

MODEM 1

U15
A5191HRTNG-XTP
MODEM
51507504-121

U6
OP2177AR
OPAMP
51506189-121

U11
MAX312LESE
SW_ANLG_SPST_NOCONN
51506339-121

TITLE	HART MODEMS 0, 1
DRAWN BY Shyam Manikanta	04-FEB-2025
DESIGNER Shyam Manikanta	04-FEB-2025
HONEYWELL INTERNATIONAL INC.	Honeywell
Honeywell Process Solutions	
HONEYWELL CONFIDENTIAL AND PROPRIETARY	
SIZE D	DRAWING NO. 50195113 ATP1
REV	
SCALE: 1:1	SHEET 17 OF 22

DxDESIGNER
20/02/2026:17:33

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MODEM 2

C61 470pF (51190142-333)
R155 215K (51192410-334)
C422 0.001uF (51190142-373)
R154 215K (51192410-334)
R157 422K (51192410-614)
R156 215K (51192410-334)
C77 220pF (51190142-297)
C72 0.001uF (51190142-373)
R158 499K (51192410-684)
R159 787K (51192410-874)
R131 3MEG (51191800-305)
R132 14.7K (51192410-173)
C63 0.01uF (51190094-493)
R133 215K (51192410-334)
GND

U22
A5191HRTNG-XTP
MODEM
51507504-121

28	TEST1	VDDA	13
29	TEST2		21
31	TEST3	VDD	30
32	TEST4		
1	TEST5	CBIAS	10
3	TEST7		
4	TEST8	CD	26
5	TEST9	TXA	7
11	TEST10	RXD	25
24	TEST11	RXAF	15
27	TEST12		
2	RESET	VSS	6
18	XIN		20
17	XOUT		12
		VSSA	19
		EP	33
22	RTS		
23	TXD		
14	RXA		
16	RXAFI		
8	AREF		
9	CDREF		

U17_P29_NC, U17_P31_NC, U17_P32_NC, U17_P24_NC, U17_P27_NC

+3V3

R160 499K (51192410-684)

HARTRST2_n (8B4)
MODEMCLK (17, 18, 8)
AREF1 (17A7, 17C7, 18A7)

HART2RTS_n (8B4) R188 280 (51192410-441)
RTS2_n
HART2CD (8B4)
HART2RXD (8B4)
C81 0.1uF (51190094-613)
R192 31.6K (51192410-493)
HART2TXD (8C4)
R189 280 (51192410-441)

C83 220pF (51190142-297)
R220 43.2K (51192410-623)
U26 OP2177AR (51506189-121)
+24V
R190 22.1K (51192410-343)
R191 100K (51192410-014)
C432 0.47uF (51190094-683)

U11
MAX312LESE
X1, X̄1[D], [S]1̄, NC, V+, GND, V-
C37 2.2uF (51506365-225)
HART2MUX (16C4)

ALL COMPONENTS ON THIS SHEET ARE NOT INSTALLED FOR TAB –235

MODEM 3

C64 470pF (51190142-333)
R162 215K (51192410-334)
C74 0.001uF (51190142-373)
R161 215K (51192410-334)
R165 422K (51192410-614)
R164 215K (51192410-334)
C80 220pF (51190142-297)
C75 0.001uF (51190142-373)
R163 499K (51192410-684)
R167 787K (51192410-874)
R134 3MEG (51191800-305)
R135 14.7K (51192410-173)
C66 0.01uF (51190094-493)
R136 215K (51192410-334)
GND

U23
A5191HRTNG-XTP
MODEM
51507504-121

28	TEST1	VDDA	13
29	TEST2		21
31	TEST3	VDD	30
32	TEST4		
1	TEST5	CBIAS	10
3	TEST7		
4	TEST8	CD	26
5	TEST9	TXA	7
11	TEST10	RXD	25
24	TEST11	RXAF	15
27	TEST12		
2	RESET	VSS	6
18	XIN		20
17	XOUT		12
		VSSA	19
		EP	33
22	RTS		
23	TXD		
14	RXA		
16	RXAFI		
8	AREF		
9	CDREF		

U18_P29_NC, U18_P31_NC, U18_P32_NC, U18_P24_NC, U18_P27_NC

+3V3

R166 499K (51192410-684)

HARTRST3_n (8B4)
MODEMCLK (17, 18, 8)
AREF1 (17A7, 17C7, 18A7)

HART3RTS_n (8B4) R193 280 (51192410-441)
RTS3_n
HART3CD (8B4)
HART3RXD (8B4)
C76 0.1uF (51190094-613)
R187 31.6K (51192410-493)
HART3TXD (8B4)
R194 280 (51192410-441)
TXD3

C78 220pF (51190142-297)
R219 43.2K (51192410-623)
U26 OP2177AR (51506189-121)
+24V
R467 22.1K (51192410-343)
R468 100K (51192410-014)
C79 0.47uF (51190094-683)

C58 15pF (51192412-150)
R110 1K (51192410-012)

U11
MAX312LESE
X1, X̄1[D], [S]1̄, NC, V+, GND, V-
C51 2.2uF (51506365-225)
HART3MUX (16A4)

HONEYWELL CONFIDENTIAL AND PROPRIETARY

DRAWN BY Shyam Manikanta	04–FEB–2025	
DESIGNER Shyam Manikanta	04–FEB–2025	

TITLE: HART MODEMS 2, 3

HONEYWELL INTERNATIONAL INC. Honeywell
Honeywell Process Solutions

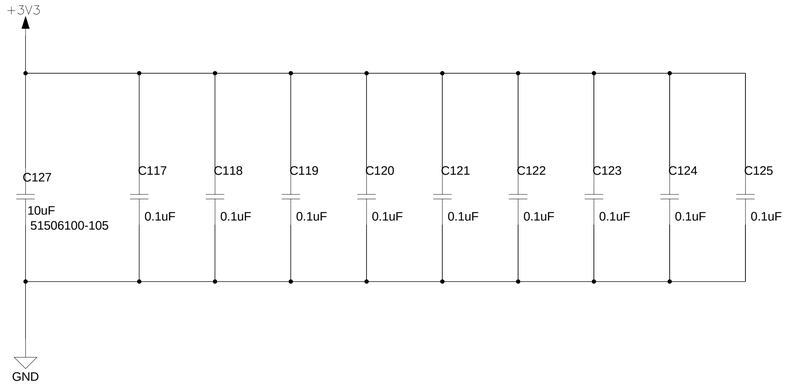
SIZE D | DRAWING NO. 50195113 | REV ATP1

SCALE: 1:1

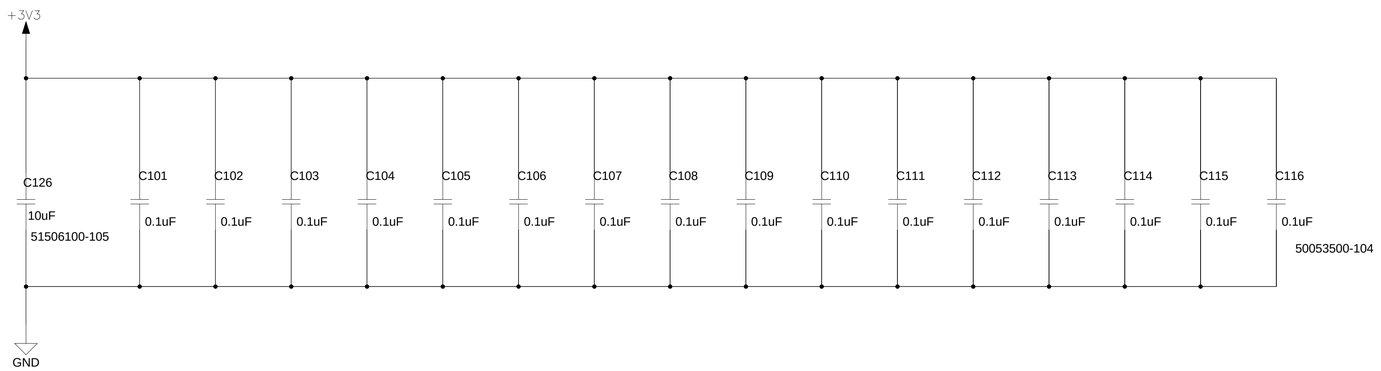
SHEET 18 OF 22

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Should be placed near VCC pins of FPGA



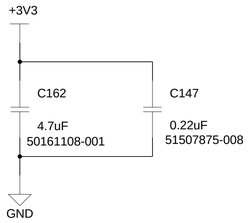
Should be placed near VCCIO pins of FPGA

DRAWN BY Shyam Manikanta	04-FEB-2025	TITLE: FPGA Decoupling Capacitors	
DESIGNER Shyam Manikanta	04-FEB-2025	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions	
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D	DRAWING NO. 50195113 ATP1 REV
		SCALE: 1:1	SHEET 19 OF 22

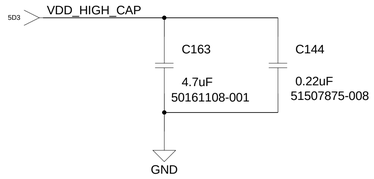
DxDESIGNER 20/02/2026:17:33

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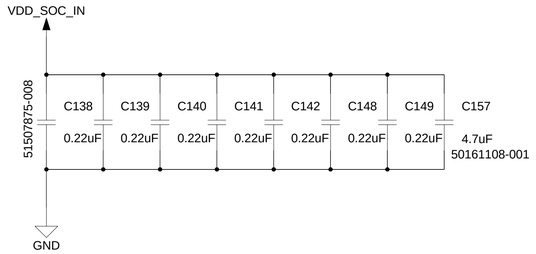
Should be placed near VDD_HIGH_IN



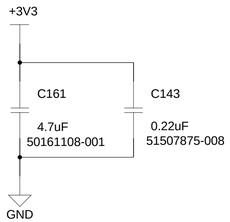
Should be placed near VDD_HIGH_CAP



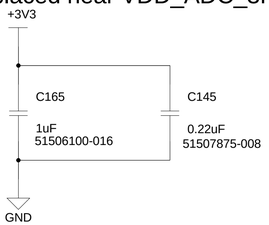
Should be placed near VDD_SOC_IN



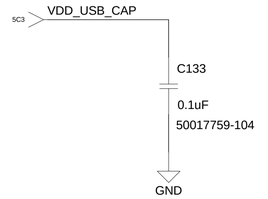
Should be placed near DCDC_IN



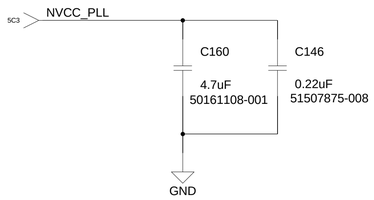
Should be placed near VDD_ADC_3P3



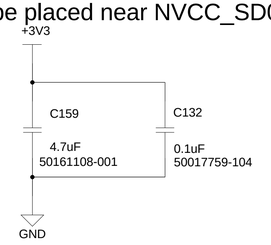
Should be placed near VDD_USB_CAP



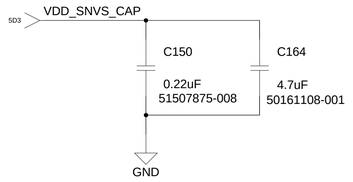
Should be placed near NVCC_PLL



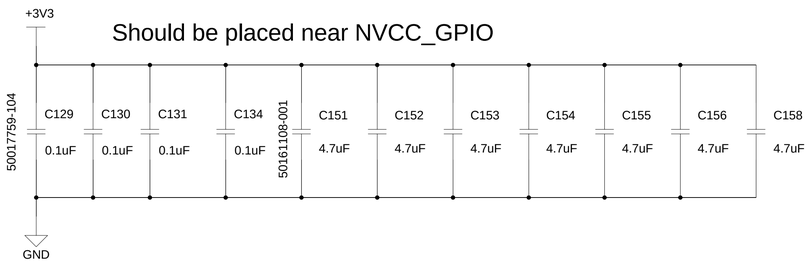
Should be placed near NVCC_SD0



Should be placed near VDD_SNVS_CAP



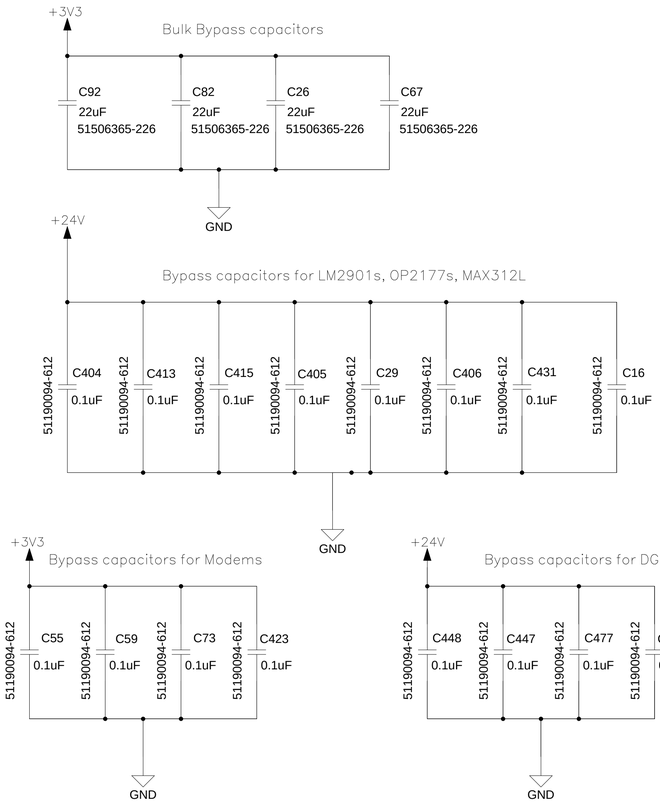
Should be placed near NVCC_GPIO



DRAWN BY Shyam Manikanta	04-FEB-2025	TITLE: uC Bypass Capacitors
DESIGNER Shyam Manikanta	04-FEB-2025	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D · DRAWING NO. 50195113 · REV ATP1
		SCALE: 1:1 · SHEET 20 OF 22

DxDESIGNER 10/04/2026:11:31

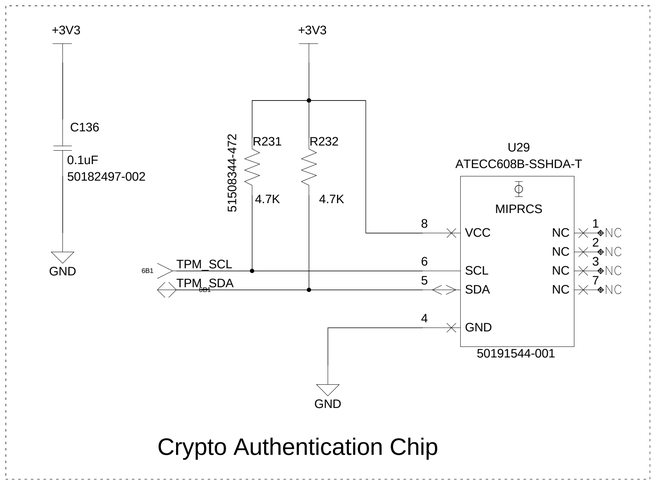
P-21



DRAWN BY Shyam Manikanta	04-FEB-2026	TITLE: Decoupling Capacitors	
DESIGNER Shyam Manikanta	04-FEB-2026	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions	
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D	DRAWING NO. 50195113 REV ATP1
		SCALE: 1:1	SHEET 21 OF 22

DxDESIGNER 01/04/2026:18:25

P-22



Crypto Authentication Chip

DRAWN BY Shyam Manikanta	04-FEB-2025	TITLE: TPM Interface
DESIGNER Shyam Manikanta	04-FEB-2025	HONEYWELL INTERNATIONAL INC. Honeywell Process Solutions
HONEYWELL CONFIDENTIAL AND PROPRIETARY		SIZE D · DRAWING NO. 50195113 · REV ATP1
		SCALE: 1:1 · SHEET 22 OF 22

DxDESIGNER 01/04/2026:11:19